Power System Cybersecurity Basics

Stealthy Cyber-attack

Strategic Cyber-defense

Model-/Data-based Cybersecurity

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Power System Cybersecurity

❑ Prerequisite

- State estimation
- Optimal power flow
- Matlab installation

❑ Content

- Stealthy Cyber-attack
- Strategic Cyber-defense
- Model-/Data-based Security

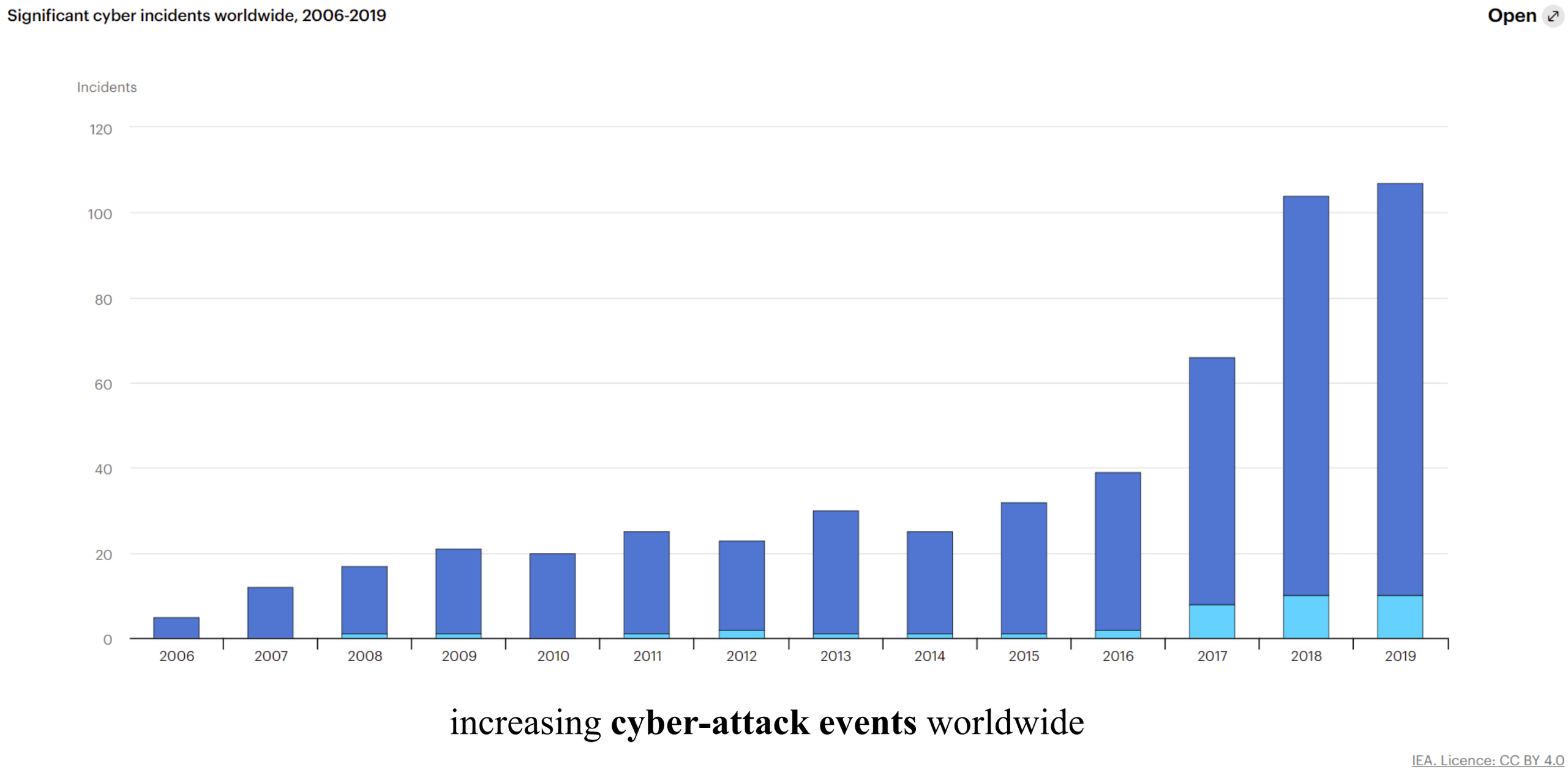
❑ Learning Outcome

- **Attacker perspective**: How the stealthy cyber-attacks bypass the bad data detection in power system state estimation.
- **System perspective**: What happen to the optimal power flow and power system security/efficiency after certain stealthy attacks.
- **Defender perspective**: How to defend the stealthy cyber-attacks.
- Similarity and difference between **model-based** and **data-driven** cybersecurity methods

Outline

- **Background**
- Attacker Perspective
  (How do cyber-attacks bypass bad data detection?)
- System Perspective
  (What would happen after stealthy cyber-attacks?)
- Defender Perspective
  (How to defend the stealthy cyber-attack?)
- Data-driven Cybersecurity
- Take-away Message

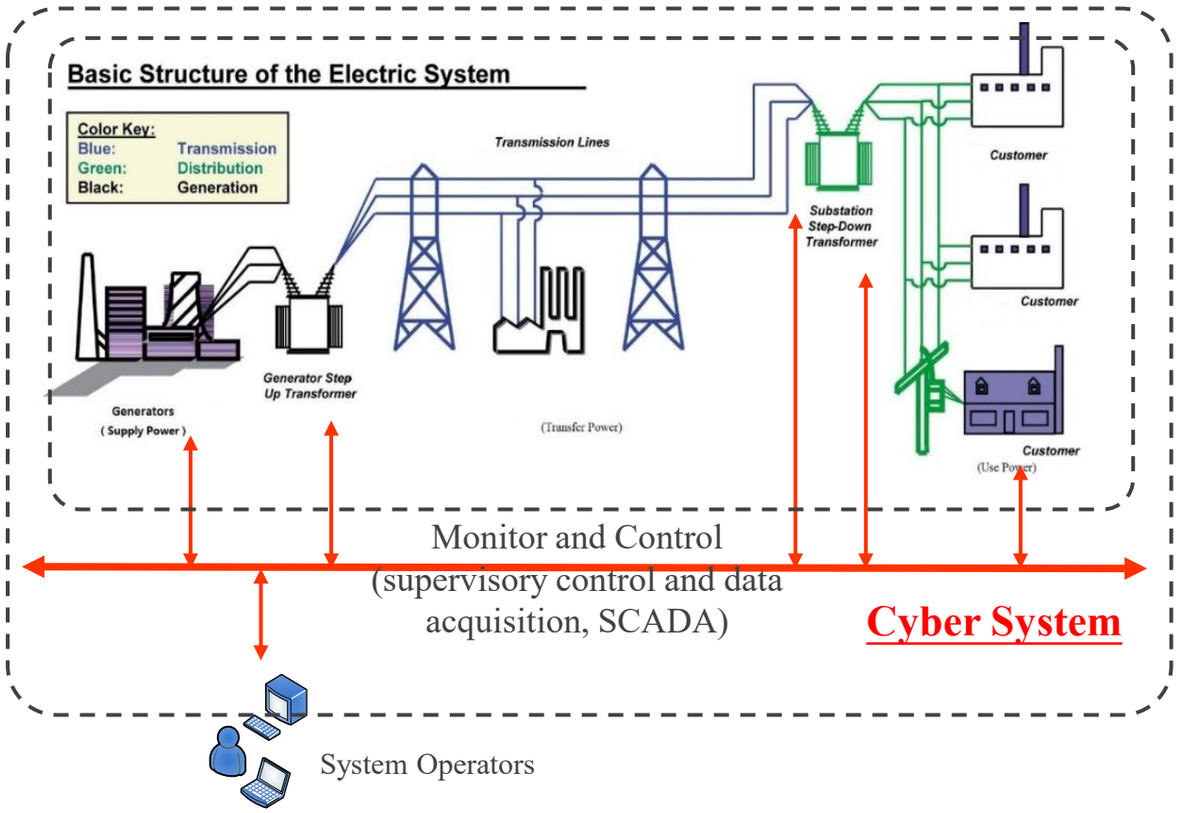
Background



increasing **cyber-attack events** worldwide

"Power systems in transition: Challenges and opportunities ahead for electricity security," International Energy Agency, Technical Report, 2020.

Background



- intrinsically cyber physical systems
- physical threats (hurricanes, floods, …)
- cyber threats (malicious cyberattacks, …)

⇩

- illegal write access of metering infrastructure
- hijack of the communication infrastructure
- unknown zero-day vulnerabilities of communication protocols
- vulnerable to cyberthreats
- unaffordable physical / economic / social impacts

Serious situation and critical importance to enhance the cybersecurity of power systems!

Background

representative **cyber-attack events** in power systems

Device	Description	Location & Time
Generation (Photovoltaic)	Around 106 MW PV project was compromised by a denial-of-service cyber-attack on the firewall.	U.S. 2021
Generation (Wind)	Around 500 MW of wind turbines was compromised by cyber-attacks on the wind operator's control system.	U.S. 2019
Prosumption (Electric Vehicle)	Researchers remotely controlled Tesla's steering system and disturbed the auto-wiper function.	China 2019
	Researchers remotely controlled a Jeep Cherokee, shunted down its acceleration, and disabled its brakes.	U.S. 2015
Prosumption (Smart Building)	Target Corporation was hacked via HVAC systems, leading to the leakage of 40 million debit and credit card accounts.	U.S. 2019
	The building management system of Google Australia Office is hacked by researchers.	Australia 2013
Prosumption	Cyber-attacks on Ukrainian power grid resulted in power outage of 225000 custmoers.	Ukraine 2015
Storage	Researchers hacked into the smart battery controller and modified battery data.	U.S. 2011

G. Liang, S. R. Weller, J. Zhao, F. Luo, and Z. Y. Dong, "The 2015 Ukraine blackout: Implications for **false data injection (FDI)** attacks," IEEE Trans. Power Syst., vol. 32, no. 4, pp. 3317–3318, Jul. 2017.

Background

- **The NIST Cybersecurity Framework 2.0**
  (National Institute of Standards and Technology) U.S.A
  C.I.A

✓ **C (confidentiality):** preserving authorized restrictions on access and disclosure, including means for protecting personal privacy and proprietary information

✓ **I (integrity):** guarding against improper information modification or destruction, and includes ensuring information non-repudiation and authenticity

✓ **A (availability):** ensuring timely and reliable access to and use of information

- **Learning outcomes exclude:**
  - Hacking
  - Malware
  - Firewall
  - Encryption/Decryption
  - How to be a good/bad guy
  - ……

- **Learning outcomes include:**
  - How the stealthy FDI cyber-attacks bypass bad data detection and then manipulate optimal power flow?
  - How the stealthy FDI cyber-attacks compromise power system security and efficiency?
  - How to defend the stealthy FDI cyber-attacks?
  - Model-based and data-driven cybersecurity?

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Recap of Power Flow Equation

linear $\boldsymbol{P} \rightarrow \boxed{\text{PF}} \rightarrow \boldsymbol{\theta}$

nonlinear $\boldsymbol{P}, \boldsymbol{Q} \rightarrow \boxed{\text{PF}} \rightarrow \boldsymbol{\theta}, \boldsymbol{V}$

$\boldsymbol{\theta}, \boldsymbol{V}, \boldsymbol{P}, \boldsymbol{Q} \in R^n \qquad \boldsymbol{p}, \boldsymbol{q} \in R^l$

Power flow equations are fully determined:

given equations / measurements (n)

= unknown state variables (n)

✓ **DC (linear) power flow (this lecture)**

	given	unknown
swing bus	$\boldsymbol{\theta}$	$\boldsymbol{P}$
other bus	$\boldsymbol{P}$	$\boldsymbol{\theta}$

$$P_i = \sum_{k=1}^{n} B_{Pik}\theta_{ik} \qquad \boldsymbol{P} = B_P\boldsymbol{\theta} \qquad B_P \in R^{n\times n}$$

$$p_{ik} = \frac{1}{X_{ik}}(\theta_i - \theta_k) \qquad \boldsymbol{p} = B_F\boldsymbol{\theta} \qquad B_F \in R^{l\times n}$$

✓ **AC (nonlinear) power flow**

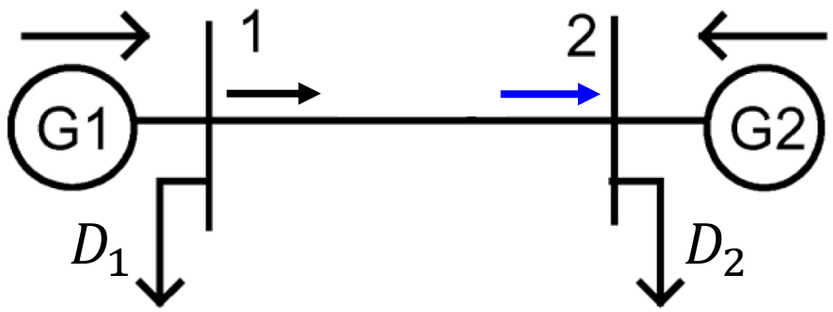
	given	unknown
swing bus	$\boldsymbol{\theta}, \boldsymbol{V}$	$\boldsymbol{P}, \boldsymbol{Q}$
PQ bus	$\boldsymbol{P}, \boldsymbol{Q}$	$\boldsymbol{\theta}, \mathbf{V}$
PV bus	$\boldsymbol{P}, \boldsymbol{V}$	$\boldsymbol{\theta}, \boldsymbol{Q}$

$$\begin{cases} P_i = V_i \sum_{k=1}^{n} V_k \left(G_{ik}\cos\theta_{ik} + B_{ik}\sin\theta_{ik}\right) \\ Q_i = V_i \sum_{k=1}^{n} V_k \left(G_{ik}\sin\theta_{ik} - B_{ik}\cos\theta_{ik}\right) \end{cases}$$

$$\begin{cases} p_{ik} = V_iV_k(G_{ik}\cos\theta_{ik} + B_{ik}\sin\theta_{ik}) - V_i^2G_{ik} \\ q_{ik} = V_iV_k(G_{ik}\sin\theta_{ik} - B_{ik}\cos\theta_{ik}) + V_i^2B_{ik} \end{cases}$$

Recap of State Estimation

A two-bus power system:



power flow equation	given $\boldsymbol{P}(+\boldsymbol{Q})$ unknown $\boldsymbol{\theta}(+\boldsymbol{V})$
state estimation measurement	$\boldsymbol{z} = [\boldsymbol{p}^T \quad \boldsymbol{P}^T]^T$ or $\boldsymbol{z} = [\boldsymbol{p}^T \quad \boldsymbol{q}^T \quad \boldsymbol{P}^T \quad \boldsymbol{Q}^T \quad \boldsymbol{V}^T]^T$
state estimation state variable	$\boldsymbol{x} = [\boldsymbol{\theta}]$ or $\boldsymbol{x} = [\boldsymbol{\theta}^T \quad \boldsymbol{V}^T]^T$

Recap of State Estimation

$$\boldsymbol{z} \longrightarrow \boxed{\text{SE}} \xrightarrow{\hat{\boldsymbol{x}}}$$

$\boldsymbol{z} \in R^m$ $\quad$ $\hat{\boldsymbol{x}} \in R^n$

State estimation equations are over determined:

given equations / measurements (m)

> unknown state variables (n)

✓ **DC (linear) power flow**

$$\boldsymbol{z} = H\boldsymbol{x} + \boldsymbol{e} \qquad H = \begin{bmatrix} B_F \\ B_P \end{bmatrix}$$

$$\boldsymbol{z} = [\boldsymbol{p}^T \quad \boldsymbol{P}^T]^T \qquad \boldsymbol{x} = [\boldsymbol{\theta}]$$

analytical solution:

$$\hat{\boldsymbol{x}} = arg \min_{\boldsymbol{x}} \|\boldsymbol{z} - H\boldsymbol{x}\|$$

$\Downarrow$ first order condition

$$\hat{\boldsymbol{x}} = (H^T H)^{-1} H^T \boldsymbol{z}$$

✓ **AC (nonlinear) power flow**

$$\boldsymbol{z} = h(\boldsymbol{x}) + \boldsymbol{e} \qquad \boldsymbol{x} = [\boldsymbol{\theta}^T \quad \boldsymbol{V}^T]^T$$

$$\boldsymbol{z} = [\boldsymbol{p}^T \quad \boldsymbol{q}^T \quad \boldsymbol{P}^T \quad \boldsymbol{Q}^T \quad \boldsymbol{V}^T]^T$$

numerical / iterative solution:

$$\hat{\boldsymbol{x}} = arg \min_{\boldsymbol{x}} \|\boldsymbol{z} - h(\boldsymbol{x})\|$$

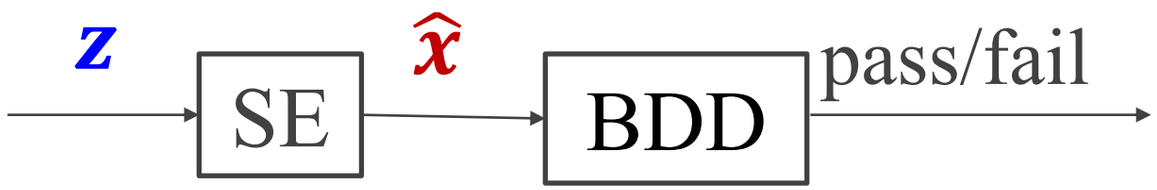
$\Downarrow$ first order condition

$$\boldsymbol{x}^{\boldsymbol{k}+1} = \boldsymbol{x}^{\boldsymbol{k}} + (K^T K)^{-1} K^T \left(z - h\left(\boldsymbol{x}^{\boldsymbol{k}}\right)\right)$$

$$K = \partial h(\boldsymbol{x}) / \partial \boldsymbol{x} \Big|_{\boldsymbol{x}^{\boldsymbol{k}}}$$

best estimation!

Residual-based Bad Data Detection in State Estimation



SE solution

$$\hat{\boldsymbol{x}} = arg \min_{\boldsymbol{x}} \|\boldsymbol{z} - H\boldsymbol{x}\| \Longrightarrow \hat{\boldsymbol{x}} = (H^T H)^{-1} H^T \boldsymbol{z}$$

residual

$$\Longrightarrow \quad r = \min_{\boldsymbol{x}} \|\boldsymbol{z} - H\boldsymbol{x}\| = \|\boldsymbol{z} - H\hat{\boldsymbol{x}}\| \text{ DC SE}$$

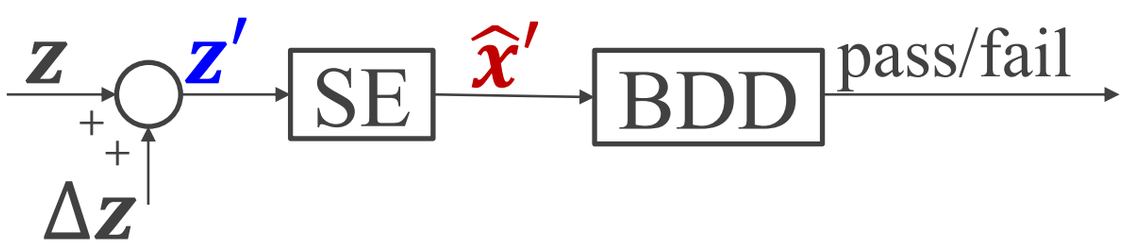
or

$$r = \min_{\boldsymbol{x}} \|\boldsymbol{z} - h(\boldsymbol{x})\| = \|\boldsymbol{z} - h(\hat{\boldsymbol{x}})\| \text{ AC SE}$$

residual test

$$\Longrightarrow \begin{cases} r \le \gamma, & \text{pass} \\ r > \gamma, & \text{fail} \end{cases}$$

False Data Injection Cyber-Attack to Bypass Bad Data Detection



$\boldsymbol{z}, \boldsymbol{z}', \Delta\boldsymbol{z} \in R^m$

$\hat{\boldsymbol{x}}, \hat{\boldsymbol{x}}', \Delta\boldsymbol{x} \in R^n$

cyber-attack injection

target states

$$\hat{\boldsymbol{x}}' = \hat{\boldsymbol{x}} + \Delta\boldsymbol{x}$$

$$\hat{\boldsymbol{x}}' = (H^T H)^{-1} H^T \boldsymbol{z}' \xrightarrow[\boldsymbol{z}' = \boldsymbol{z} + \Delta\boldsymbol{z}]{} \Delta\boldsymbol{x} = (H^T H)^{-1} H^T \Delta\boldsymbol{z}$$

consequent residual after cyber-attack injection:

$$r' = \|\boldsymbol{z}' - H\hat{\boldsymbol{x}}'\| = \|(\boldsymbol{z} - \boldsymbol{H}\hat{\boldsymbol{x}}) + (\Delta\boldsymbol{z} - H(H^T H)^{-1} H^T \Delta\boldsymbol{z})\|$$

If the cyber-attack injection is designed as $\Delta\boldsymbol{z} = H\Delta\boldsymbol{x}$

Then: $r' = r \leq \gamma$ **FDI attacks successfully bypass the BDD !**

Y. Liu, P. Ning, and M. K. Reiter, "False data injection attacks against state estimation in electric power grids," *ACM Trans. Inf. Syst. Security*, vol. 14, no. 1, pp. 1–32, May 2011.

A Specific FDI Cyber-attack
(load redistribution cyber-attack)

FDI cyber-attack:
(H **implies power flow equation**)

$$\Delta \boldsymbol{z} = H \Delta \boldsymbol{x}$$

$$\begin{bmatrix} \Delta \boldsymbol{p} \\ \Delta \boldsymbol{P} \end{bmatrix} = \begin{bmatrix} B_F \\ B_P \end{bmatrix} \Delta \boldsymbol{\theta}$$

Power flow equation: $\boldsymbol{p} = \boldsymbol{S}\,\boldsymbol{P} = \boldsymbol{S}\,(\boldsymbol{U}\,\boldsymbol{G} - \boldsymbol{V}\,\boldsymbol{D})$ $\quad \boldsymbol{S} = B_F {B_P}^{-1}$

Assumption:

- Load measurements are attackable within certain ranges.
- Generation measurement is hard to attack, yielding $\Delta \boldsymbol{G} = \boldsymbol{0}$.

Load redistribution cyber-attack:

$$\underline{\Delta \boldsymbol{D}} \leq \Delta \boldsymbol{D} \leq \overline{\Delta \boldsymbol{D}}$$

$$\boldsymbol{1}^T \Delta \boldsymbol{D} = 0$$

$$\Delta \boldsymbol{p} = -\boldsymbol{S}\,\boldsymbol{V}\,\Delta \boldsymbol{D}$$

⇨ **Criterion to design a stealthy cyber-attack!**

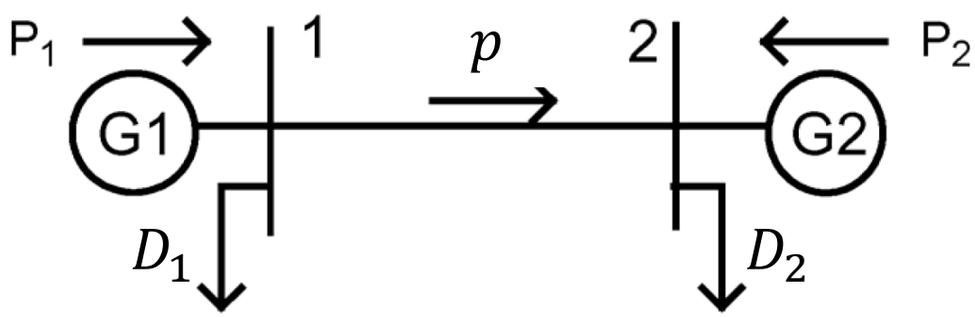
Why FDI cyber-attack?

- Previous attacks not focus on power systems, require too strong assumption.
- Based on power flow equation / state estimation.
- Measurement data rather than command data.

Yuan, Y., Li, Z., & Ren, K. (2011). Modeling load redistribution attacks in power systems. IEEE Transactions on Smart Grid, 2(2), 382-390.

Example 4.1: Two-bus Load Redistribution Attack

A two-bus power system:



General criterion to be stealthy:

$$\mathbf{1}^T \Delta \boldsymbol{D} = 0$$

$$\underline{\Delta \boldsymbol{D}} \leq \Delta \boldsymbol{D} \leq \overline{\Delta \boldsymbol{D}}$$

$$\Delta \boldsymbol{p} = -\boldsymbol{S}\,\boldsymbol{V}\,\Delta \boldsymbol{D}$$

$$\boldsymbol{S} = [0 \quad -1] \qquad \mathbf{V} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Stealthy cyber-attack on the two-bus system:

$$\Delta D_1 + \Delta D_2 = 0$$

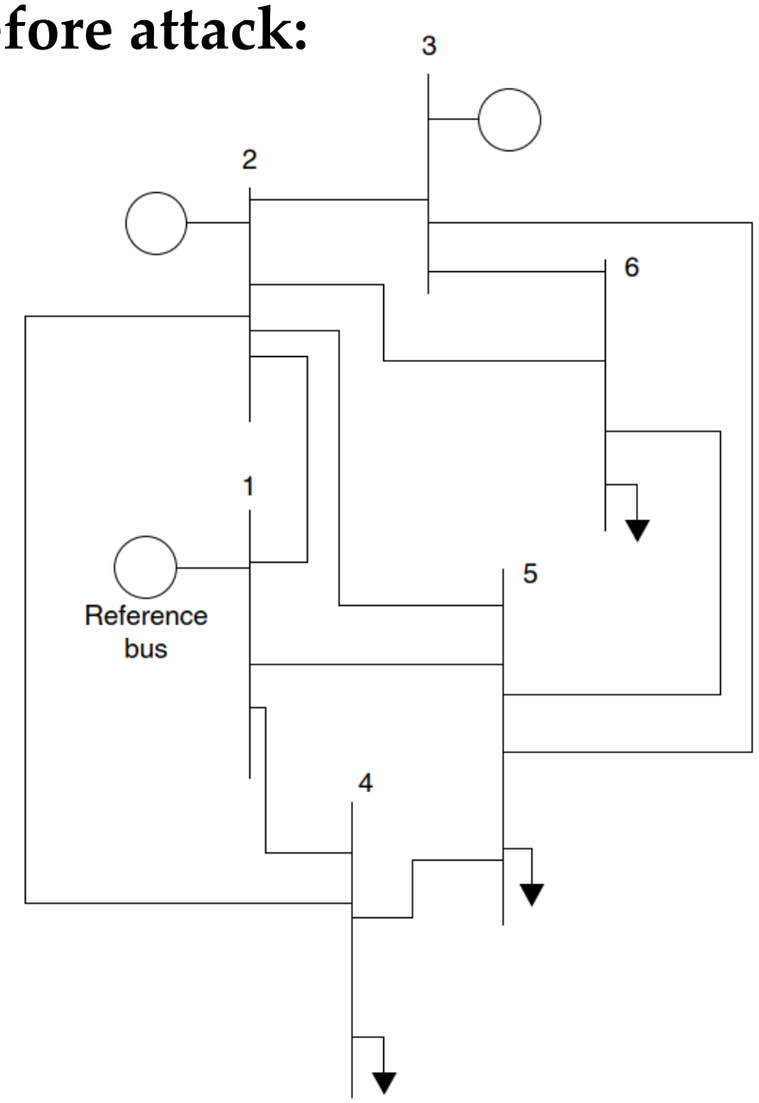
$$-\tau \begin{bmatrix} D_1 \\ D_2 \end{bmatrix} \leq \begin{bmatrix} \Delta D_1 \\ \Delta D_2 \end{bmatrix} \leq \tau \begin{bmatrix} D_1 \\ D_2 \end{bmatrix}$$

$$\Delta p = -[0 \quad -1] \begin{bmatrix} \Delta D_1 \\ \Delta D_2 \end{bmatrix}$$

Yuan, Y. (2012). Load redistribution attacks and protection strategy design in electric power systems. Illinois Institute of Technology.

Example 4.2: 6-bus Load Redistribution Attack

Before attack:



Full measurement

```
 ideal meas      noisy meas
ans =

   2.5328e-01    2.5362e-01
   4.1567e-01    4.1555e-01
   3.3104e-01    3.3185e-01
   1.8537e-02    1.8469e-02
   3.2478e-01    3.2371e-01
   1.6219e-01    1.6300e-01
   2.4778e-01    2.4590e-01
   1.6932e-01    1.6906e-01
   4.4922e-01    4.4969e-01
   4.0448e-02    4.0768e-02
   2.9986e-03    3.2413e-03
   1.0000e+00    9.9990e-01
   5.0000e-01    4.9955e-01
   6.0000e-01    5.9959e-01
  -7.0000e-01   -6.9850e-01
  -7.0000e-01   -7.0013e-01
  -7.0000e-01   -6.9773e-01
```

$$z = [p^T \quad P^T]^T$$

line flow measurement

bus injection measurement

Example 4.2: 6-bus Load Redistribution Attack

Before attack:

```
 ideal meas      noisy meas
ans =

   2.5328e-01   2.5362e-01
   4.1567e-01   4.1555e-01
   3.3104e-01   3.3185e-01
   1.8537e-02   1.8469e-02
   3.2478e-01   3.2371e-01
   1.6219e-01   1.6300e-01
   2.4778e-01   2.4590e-01
   1.6932e-01   1.6906e-01
   4.4922e-01   4.4969e-01
   4.0448e-02   4.0768e-02
   2.9986e-03   3.2413e-03
   1.0000e+00   9.9990e-01
   5.0000e-01   4.9955e-01
   6.0000e-01   5.9959e-01
  -7.0000e-01  -6.9850e-01
  -7.0000e-01  -7.0013e-01
  -7.0000e-01  -6.9773e-01
```

$$\boldsymbol{x} = [\boldsymbol{\theta}] \quad \boldsymbol{z} = [\boldsymbol{p}^T \quad \boldsymbol{P}^T]^T \quad H = \begin{bmatrix} B_F \\ B_P \end{bmatrix}$$

$$\hat{\boldsymbol{x}} = (H^T H)^{-1} H^T \boldsymbol{z}$$

```
 ideal state   estimated state
ans =

             0              0
  -2.9024e+00  -2.9026e+00
  -3.1679e+00  -3.1669e+00
  -4.7632e+00  -4.7597e+00
  -5.6902e+00  -5.6908e+00
  -5.7418e+00  -5.7366e+00
```

$$r' = \|\boldsymbol{z}' - H\hat{\boldsymbol{x}}'\| = \|(\boldsymbol{z} - \boldsymbol{H}\hat{\boldsymbol{x}}) + (\Delta\boldsymbol{z} - H(H^T H)^{-1} H^T \Delta\boldsymbol{z})\|$$

```
LNR_good =

   2.7086e-03
```

Example 4.2: 6-bus Load Redistribution Attack

After attack:

```
delta_x = [0; 0; 0; 0; 0; 10 * pi / 180];
```

$$\begin{bmatrix} \Delta \boldsymbol{p} \\ \Delta \boldsymbol{P} \end{bmatrix} = \begin{bmatrix} B_F \\ B_P \end{bmatrix} \Delta \boldsymbol{\theta}$$

```
 ideal meas      noisy meas   attack inject     compromised meas
ans =

   2.5328e-01    2.5362e-01              0    2.5362e-01
   4.1567e-01    4.1555e-01              0    4.1555e-01
   3.3104e-01    3.3185e-01              0    3.3185e-01
   1.8537e-02    1.8469e-02              0    1.8469e-02
   3.2478e-01    3.2371e-01              0    3.2371e-01
   1.6219e-01    1.6300e-01              0    1.6300e-01
   2.4778e-01    2.4590e-01  -8.7266e-01   -6.2676e-01
   1.6932e-01    1.6906e-01              0    1.6906e-01
   4.4922e-01    4.4969e-01  -1.7453e+00   -1.2956e+00
   4.0448e-02    4.0768e-02              0    4.0768e-02
   2.9986e-03    3.2413e-03  -5.8178e-01   -5.7854e-01
   1.0000e+00    9.9990e-01              0    9.9990e-01
   5.0000e-01    4.9955e-01  -8.7266e-01   -3.7312e-01
   6.0000e-01    5.9959e-01  -1.7453e+00   -1.1457e+00
  -7.0000e-01   -6.9850e-01              0   -6.9850e-01
  -7.0000e-01   -7.0013e-01  -5.8178e-01   -1.2819e+00
  -7.0000e-01   -6.9773e-01   3.1998e+00    2.5020e+00
```

No need to inject all the measurement.

line flow measurement

bus injection measurement

$$\Delta \boldsymbol{z} = H \Delta \boldsymbol{x}$$

Example 4.2: 6-bus Load Redistribution Attack

After attack:

```
delta_x = [0; 0; 0; 0; 0; 10 * pi / 180];
```

Δx

```
 ideal state   before-attack estimated state   after-attack estimated state
ans =

             0             0             0
  -2.9024e+00  -2.9026e+00  -2.9026e+00
  -3.1679e+00  -3.1669e+00  -3.1669e+00
  -4.7632e+00  -4.7597e+00  -4.7597e+00
  -5.6902e+00  -5.6908e+00  -5.6908e+00
  -5.7418e+00  -5.7366e+00  4.2634e+00
```

Attack target achieved!

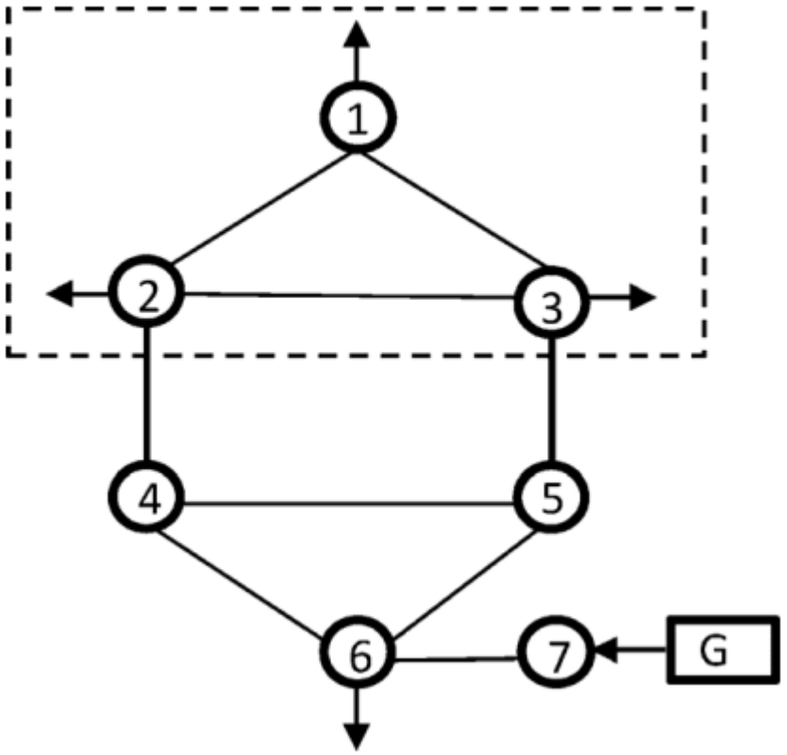
```
 before-attack residual   after-attack residual
ans =

   2.7086e-03   2.6444e-03
```

Stealthy attack: manage to bypass BDD!

$$r' = r \leq \gamma$$

Advanced FDI Cyber-Attack

PF —over determined equations→ SE —pass the BDD→ FDI



✓ **information requirements for original FDI attacks**

1. The whole topology and admittance information of AC grid
2. Write access of all the power injection / power flow sensors
3. Read access of several power injection / power flow sensors

✓ **existing researches of FDI attacks**

Criticism: Original FDI attack needs too many costs, and has unclear benefits.
(Rome was not built in a day.)

1. Information requirements of FDI: standard / local / blind
2. Least-cost attack: l_0-norm optimization, compressive sensing
3. Worst-security/efficiency cyber-attack

Liu, X., & Li, Z. (2014). Local load redistribution attacks in power systems with incomplete network information. IEEE Transactions on Smart Grid, 5(4), 1665-1676.

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  (How do cyber-attacks bypass bad data detection?)
- **System Perspective**
  (What would happen after stealthy cyber-attacks?)
- Defender Perspective
  (How to defend the stealthy cyber-attack?)
- Data-driven Cybersecurity
- Take-away Message

OPF influenced by FDI cyber-attack

DC OPF before stealthy cyber-attacks:
(every 5-15 minutes)

$$\min_{\boldsymbol{G}} \boldsymbol{c}^T \boldsymbol{G}$$

$$\text{s.t. } \mathbf{1}^T \boldsymbol{D} = \mathbf{1}^T \boldsymbol{G}$$

$$\boldsymbol{p} = \boldsymbol{S}(\boldsymbol{U}\,\boldsymbol{G} - \boldsymbol{V}\,\boldsymbol{D})$$

$$\underline{\boldsymbol{G}} \leq \boldsymbol{G} \leq \overline{\boldsymbol{G}}$$

$$\underline{\boldsymbol{p}} \leq \boldsymbol{p} \leq \overline{\boldsymbol{p}}$$

OPF after stealthy cyber-attacks:

$$\min_{\boldsymbol{G}} \boldsymbol{c}^T \boldsymbol{G}$$

$$\text{s.t. } \mathbf{1}^T \boldsymbol{D} = \mathbf{1}^T \boldsymbol{G}$$

$$\boldsymbol{p}' = \left(\boldsymbol{S}\boldsymbol{U}\,\boldsymbol{G} - \boldsymbol{V}\,(\boldsymbol{D} + \Delta\boldsymbol{D})\right)$$

$$= \boldsymbol{p} + \Delta\boldsymbol{p}$$

$$\underline{\boldsymbol{G}} \leq \boldsymbol{G} \leq \overline{\boldsymbol{G}} \qquad \underline{\boldsymbol{p}} \leq \boldsymbol{p}' \leq \overline{\boldsymbol{p}}$$

Generation **re**-dispatch!
(inappropriate decision!)

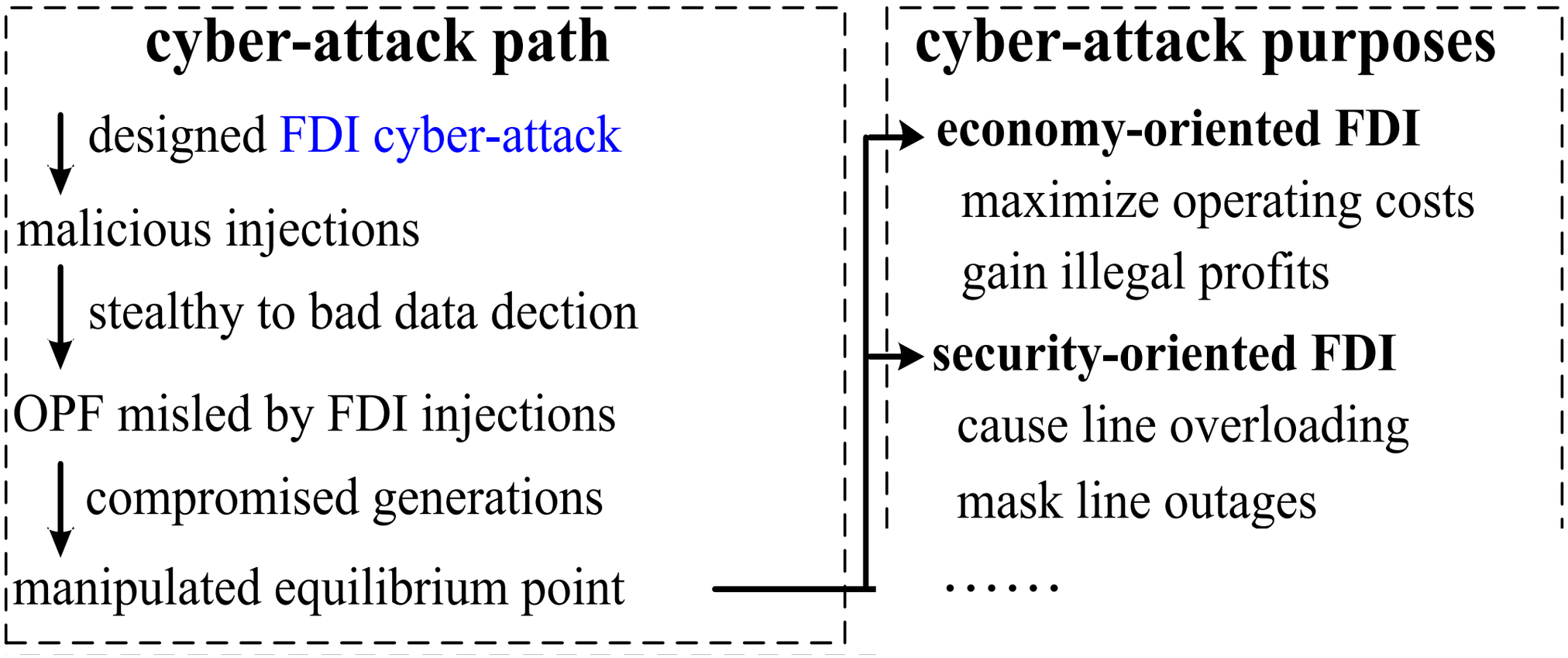
General criterion to be stealthy:

$$\mathbf{1}^T \Delta\boldsymbol{D} = 0$$

$$\underline{\Delta\boldsymbol{D}} \leq \Delta\boldsymbol{D} \leq \overline{\Delta\boldsymbol{D}}$$

$$\Delta\boldsymbol{p} = -\boldsymbol{S}\,\boldsymbol{V}\,\Delta\boldsymbol{D}$$

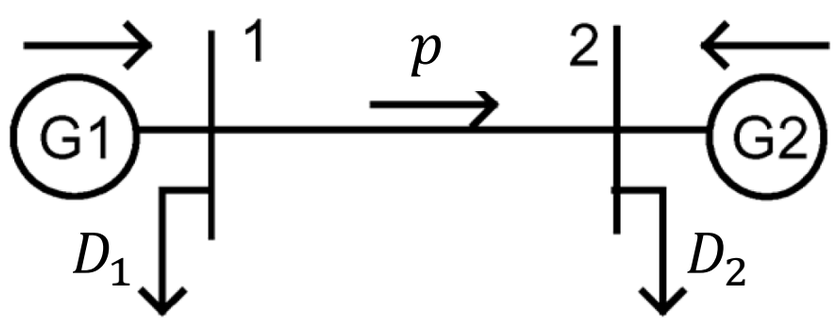
OPF influenced by FDI cyber-attack



J. Hou, J. Wang, Y. Song, W. Sun, and Y. Hou, “Small-signal angle stability-oriented false data injection cyber-attacks on power systems,” IEEE Trans. Smart Grid, vol. 14, no. 1, pp. 635–648, Jan. 2023.

Example 4.3: Load Redistribution Attack

A two-bus power system:



Generation cost:

$c_1 = 10$ \$/pu $c_2 = 20$ \$/pu

Load:

$D_1 = D_2 = 20$ pu

Line flow limit:

-5 pu $\leq p \leq$ 5 pu

Original OPF (**before** attack):

$G_1 = 25$ pu $G_1 = 15$ pu

$p = 5$ pu

Fully make use of the line flow limit to minimize the generation cost!

Generation cost:

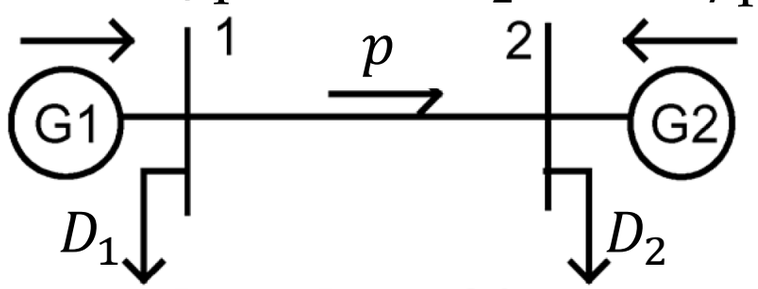
$f = c_1 G_1 + c_2 G_2 = 550$ \$

Yuan, Y. (2012). Load redistribution attacks and protection strategy design in electric power systems. Illinois Institute of Technology.

Example 4.3: Load Redistribution Attack (**security**-oriented)

$-5 \text{ pu} \le p \le 5 \text{ pu}$

$c_1 = 10 \text{ \$/pu}$ $\quad$ $c_2 = 20 \text{ \$/pu}$



$D_1 = D_2 = 20 \text{ pu}$

Criterion to be stealthy:

$$\Delta D_1 + \Delta D_2 = 0$$

$$\Delta p = -[0 \quad -1]\begin{bmatrix}\Delta D_1 \\ \Delta D_2\end{bmatrix}$$

$$-\tau\begin{bmatrix}D_1 \\ D_2\end{bmatrix} \le \begin{bmatrix}\Delta D_1 \\ \Delta D_2\end{bmatrix} \le \tau\begin{bmatrix}D_1 \\ D_2\end{bmatrix}$$

	before-attack OPF	① LR attack injection	③ after-attack OPF
D_1 pu	20		20
D_2 pu	20		20
G_1 pu	25		**27**
G_2 pu	15		**13**
p pu	5		**7**
f \$	550		530
ΔD_1 pu	0	**2** ①	2
ΔD_2 pu	0	**-2** ①	-2
Δp pu	0	**-2** ①	-2
$D_1^{\text{screen}} = D_1 + \Delta D_1$	20	22 ②	22
$D_2^{\text{screen}} = D_2 + \Delta D_2$	20	18 ②	18
$p^{\text{screen}} = p + \Delta p$	5	3	**5**

③ "OPF again!"

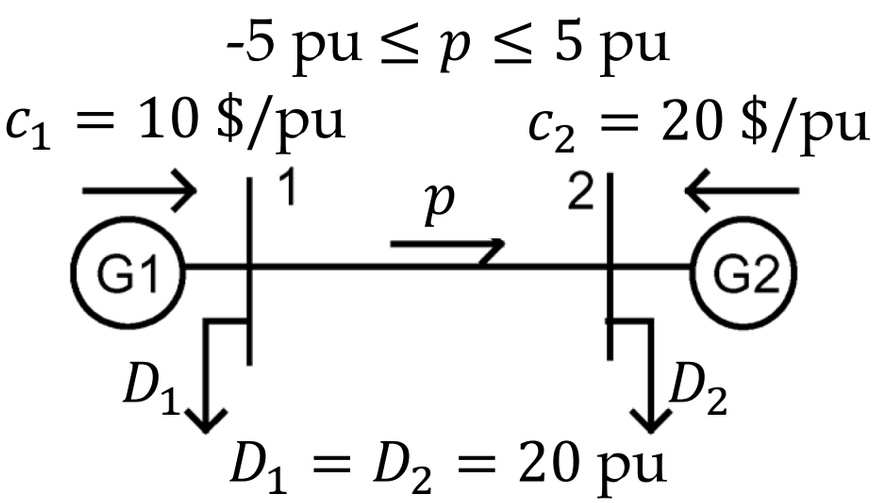
⑤ **But, line overloading!**

④ "After OPF again, I save money!"

④ "After OPF again, **everything looks fine** on the monitor screen."

② "Wow, additional line flow margin! Let's OPF again to save money!"

Example 4.3: Load Redistribution Attack (**security**-oriented)



Security-oriented LR attack:

- **On the screen**: Everything looks fine!
- **In fact**: Line overloading right now (by saving money)!

⇩

Insight: The lines close to limits are cyber vulnerable.

Criterion to be stealthy:

$$\Delta D_1 + \Delta D_2 = 0$$

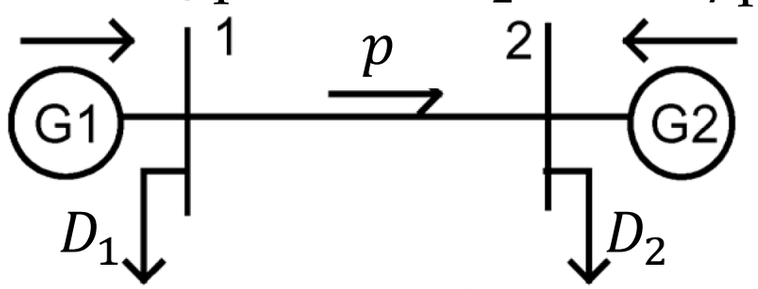
$$\Delta p = -[0 \quad -1]\begin{bmatrix}\Delta D_1 \\ \Delta D_2\end{bmatrix}$$

$$-\tau\begin{bmatrix}D_1 \\ D_2\end{bmatrix} \leq \begin{bmatrix}\Delta D_1 \\ \Delta D_2\end{bmatrix} \leq \tau\begin{bmatrix}D_1 \\ D_2\end{bmatrix}$$

Example 4.4: Load Redistribution Attack (**efficiency**-oriented)

$-5 \text{ pu} \le p \le 5 \text{ pu}$

$c_1 = 10 \text{ \$/pu}$ $\quad$ $c_2 = 20 \text{ \$/pu}$



$D_1 = D_2 = 20 \text{ pu}$

Criterion to be stealthy:

$$\Delta D_1 + \Delta D_2 = 0$$

$$\Delta p = -[0 \quad -1]\begin{bmatrix}\Delta D_1 \\ \Delta D_2\end{bmatrix}$$

$$-\tau\begin{bmatrix}D_1 \\ D_2\end{bmatrix} \le \begin{bmatrix}\Delta D_1 \\ \Delta D_2\end{bmatrix} \le \tau\begin{bmatrix}D_1 \\ D_2\end{bmatrix}$$

	before-attack OPF	① → LR attack injection	③ → after-attack OPF
D_1 pu	20		20
D_2 pu	20		20
G_1 pu	25		23
G_2 pu	15		17
p pu	5		3
f \$	550		**570**
ΔD_1 pu	0	**-2** ①	-2
ΔD_2 pu	0	**2** ①	2
Δp pu	0	**2** ①	2
$D_1^{\text{screen}} = D_1 + \Delta D_1$	20	18 ②	18
$D_2^{\text{screen}} = D_2 + \Delta D_2$	20	22 ②	22
$p^{\text{screen}} = p + \Delta p$	5	7	5

③ "OPF again!"

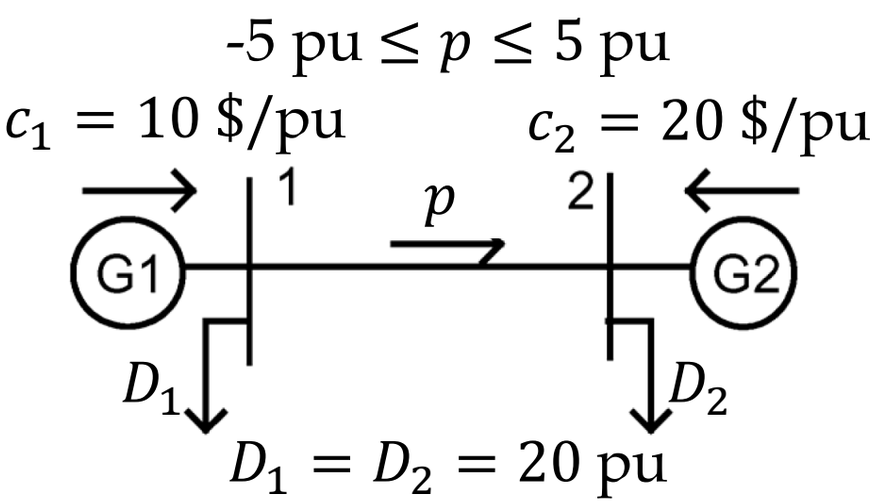
Does not fully make ⑤ use of line limit!

⑤ **Lose money / efficiency!**

"After OPF again, I ④ save an overloaded line."

② "Oops, line overloading found! Let's OPF again to save the line!"

Example 4.4: Load Redistribution Attack (**efficiency**-oriented)



Criterion to be stealthy:

$$\Delta D_1 + \Delta D_2 = 0$$

$$\Delta p = -[0 \quad -1]\begin{bmatrix}\Delta D_1 \\ \Delta D_2\end{bmatrix}$$

$$-\tau\begin{bmatrix}D_1 \\ D_2\end{bmatrix} \leq \begin{bmatrix}\Delta D_1 \\ \Delta D_2\end{bmatrix} \leq \tau\begin{bmatrix}D_1 \\ D_2\end{bmatrix}$$

Efficiency-oriented LR attack:

- **On the screen**: I just saved an overloaded line!
- **In fact**: Lose money/efficiency (by avoiding overloading)!

⇩

Insight: If the lines interconnected to the cheap generators are attacked, the expensive generators would output more.

Outline

- Background
- Attacker Perspective
  (How do cyber-attacks bypass bad data detection?)
- System Perspective
  (What would happen after stealthy cyber-attacks?)
- **Defender Perspective**
  (How to defend the stealthy cyber-attack?)
- Data-driven Cybersecurity
- Take-away Message

Cyber-defense Strategy

General criterion to be stealthy (**without** defense):

$$\mathbf{1}^T \Delta \boldsymbol{D} = 0 \qquad \underline{\Delta \boldsymbol{D}} \leq \Delta \boldsymbol{D} \leq \overline{\Delta \boldsymbol{D}} \qquad \Delta \boldsymbol{p} = -\boldsymbol{S}\,\boldsymbol{V}\,\Delta \boldsymbol{D}$$

General criterion to be stealthy (**with** defense):

generalize ⇓ ⇑ special case

$$\mathbf{1}^T \Delta \boldsymbol{D} = 0 \qquad (\mathbf{1} - \boldsymbol{\delta}) \cdot \underline{\Delta \boldsymbol{D}} \leq \Delta \boldsymbol{D} \leq (\mathbf{1} - \boldsymbol{\delta}) \cdot \overline{\Delta \boldsymbol{D}}$$

$$-(\mathbf{1} - \boldsymbol{\rho}) M \leq \Delta \boldsymbol{p} = -\boldsymbol{S}\,\boldsymbol{V}\,\Delta \boldsymbol{D} \leq (\mathbf{1} - \boldsymbol{\rho}) M$$

$\delta \in \{0,1\}$ cyber-defense decision in loads (1 indicates with defense, 0 indicate without defense)

$\rho \in \{0,1\}$ cyber-defense decision in line flows

M sufficiently large positive constant

C_δ, C_ρ cyber-defense cost in loads/lines

$\cdot$ element-wise product

Cyber-defense Strategy

1 spatial- / rule- based strategy

1. standard largest normalized residual
2. minimal set of meters $\boldsymbol{z}$
3. minimal set of state variables $\boldsymbol{x}$ (PMU)
4. cross validation of extended sensors (based on physical laws)

Cyber-defense Strategy

- One index to evaluate the defense strategy:

  Maximum injection of the l-th line:

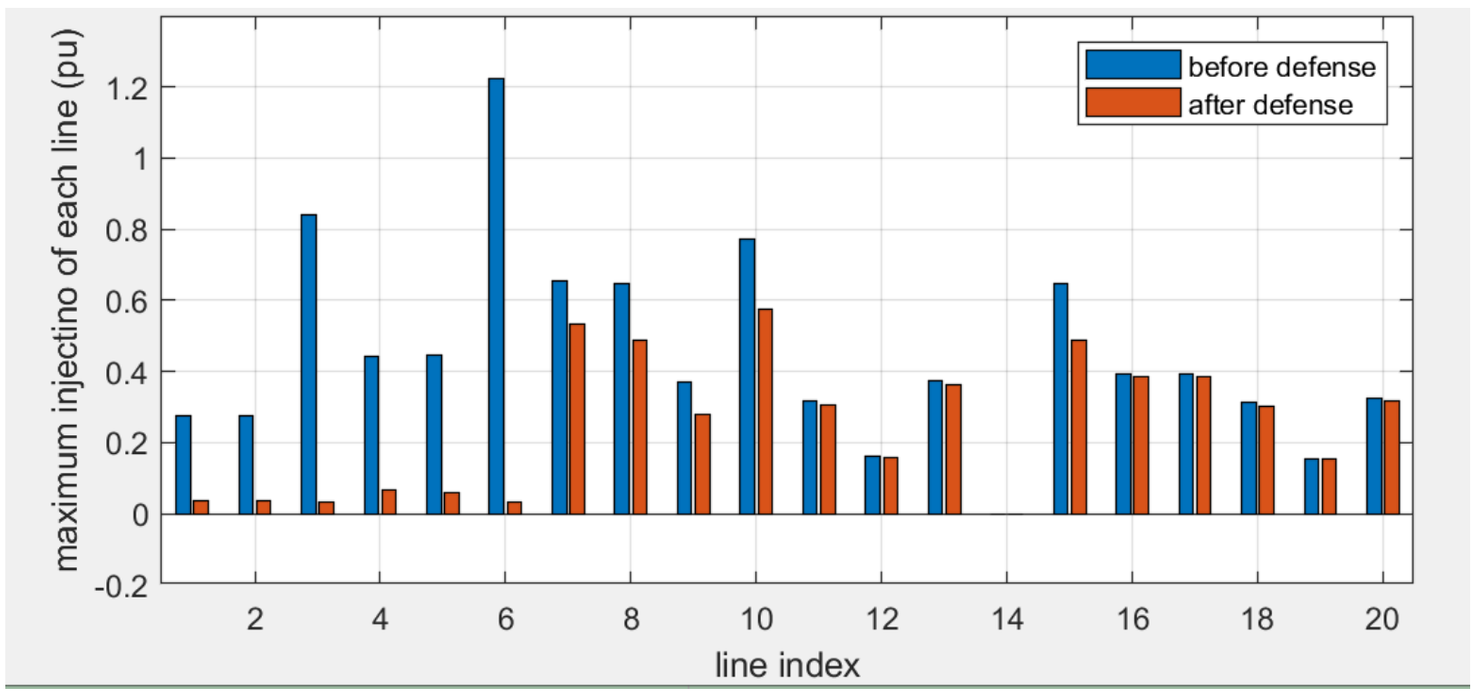
$$\max_{\Delta \boldsymbol{D}, \Delta \boldsymbol{p}} \Delta p_l / \overline{p_l}$$

Attack-induced region: $\sum_l \Delta p_l / \overline{p_l}$

The smaller the attack-induced region, the safer the power system!

$$\text{s.t.} \quad \mathbf{1}^T \Delta \boldsymbol{D} = 0 \qquad (\mathbf{1}^T - \boldsymbol{\delta}^T)\, \underline{\Delta \boldsymbol{D}} \leq \Delta \boldsymbol{D} \leq (\mathbf{1}^T - \boldsymbol{\delta}^T)\, \overline{\Delta \boldsymbol{D}}$$

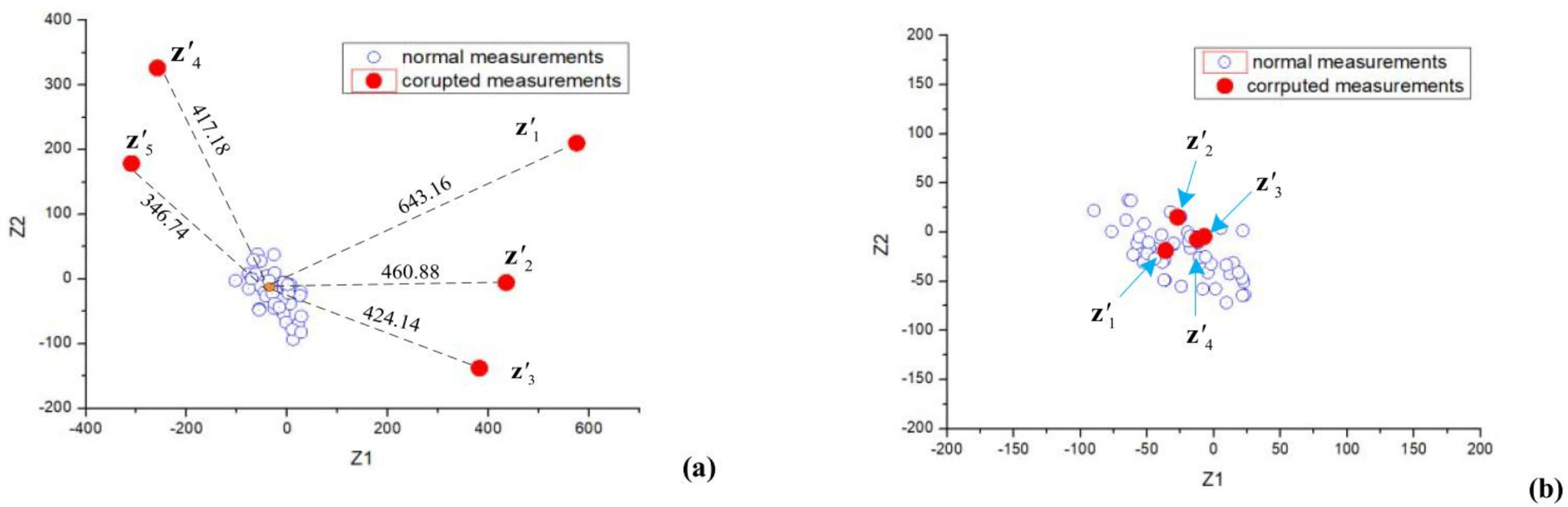
$$-(\mathbf{1}^T - \boldsymbol{\rho}^T)\, M \leq \Delta \boldsymbol{p} = -\boldsymbol{S}\, \boldsymbol{V}\, \Delta \boldsymbol{D} \leq (\mathbf{1}^T - \boldsymbol{\rho}^T)\, M$$



Cyber-defense Strategy

2 temporal- / data- based strategy (rely on training data)

1. comparison of measurements and predicted values
   (based on historical data)
2. feature extraction
   (PCA, tensor decomposition, KL-divergence, RNN)
3. supervised / unsupervised learning (classification)
4. hypothesis test (statistical technique)



Liu, X., Song, Y., & Li, Z. (2019). Dummy data attacks in power systems. IEEE Transactions on Smart Grid, 11(2), 1792-1795.

Cyber-defense Strategy

3 active

moving target defense

(graph theory, complex network)

$$\Delta \boldsymbol{z} = H \Delta \boldsymbol{x} \neq H' \Delta \boldsymbol{x}$$

4 computer science technique

firewall, encryption, abnormal data flow

5 cyber security technique (industrial security)

physical watermark

Outline

- Background
- Attacker Perspective
  (How do cyber-attacks bypass bad data detection?)
- System Perspective
  (What would happen after stealthy cyber-attacks?)
- Defender Perspective
  (How to defend the stealthy cyber-attack?)
- **Data-driven Cybersecurity**
- Take-away Message

Model-based & Data-based OPF

$$\min_{\boldsymbol{G}} \boldsymbol{c}^T \boldsymbol{G}$$

$$\text{s.t.} \quad \mathbf{1}^T \boldsymbol{D} = \mathbf{1}^T \boldsymbol{G}$$

$$\boldsymbol{p} = \boldsymbol{S}(\boldsymbol{U}\,\boldsymbol{G} - \boldsymbol{V}\,\boldsymbol{D})$$

$$\underline{\boldsymbol{G}} \leq \boldsymbol{G} \leq \overline{\boldsymbol{G}}$$

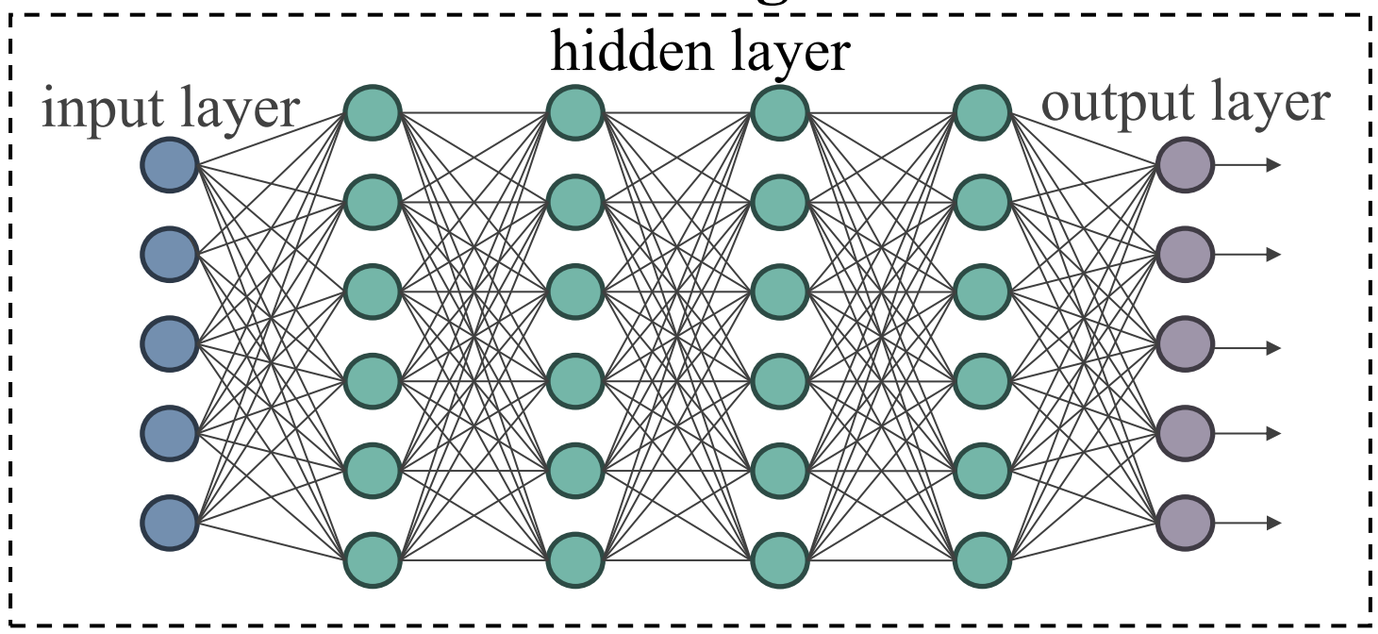
$$\underline{\boldsymbol{p}} \leq \boldsymbol{p} \leq \overline{\boldsymbol{p}}$$

Input: power consumption $\boldsymbol{D}$

model-based OPF using solver

Output: power generation $\boldsymbol{G}$

data-driven OPF using neural network



Training Data-based OPF

Model-based OPF:

- **Model** (power flow equation, bad data detection)
- **Attack** (false data injection attack)
- **Defend** (passive/active)

Data-based OPF:

- **Train** (neural network, min loss function)
- **Attack** (adversarial attack)
- **Defend** (passive/active)

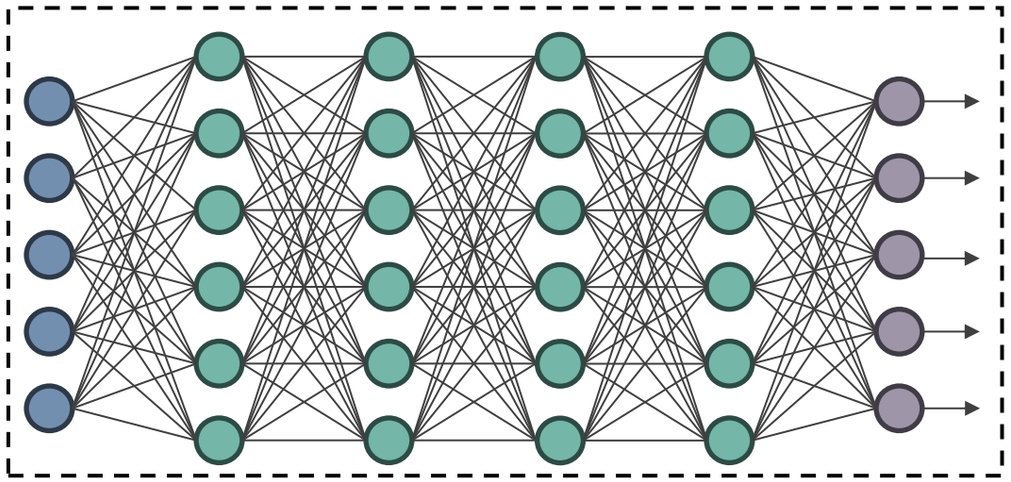
$$\min_{\boldsymbol{G}} \boldsymbol{c}^T \boldsymbol{G}$$

$$\text{s.t.} \quad \mathbf{1}^T \boldsymbol{D} = \mathbf{1}^T \boldsymbol{G}$$

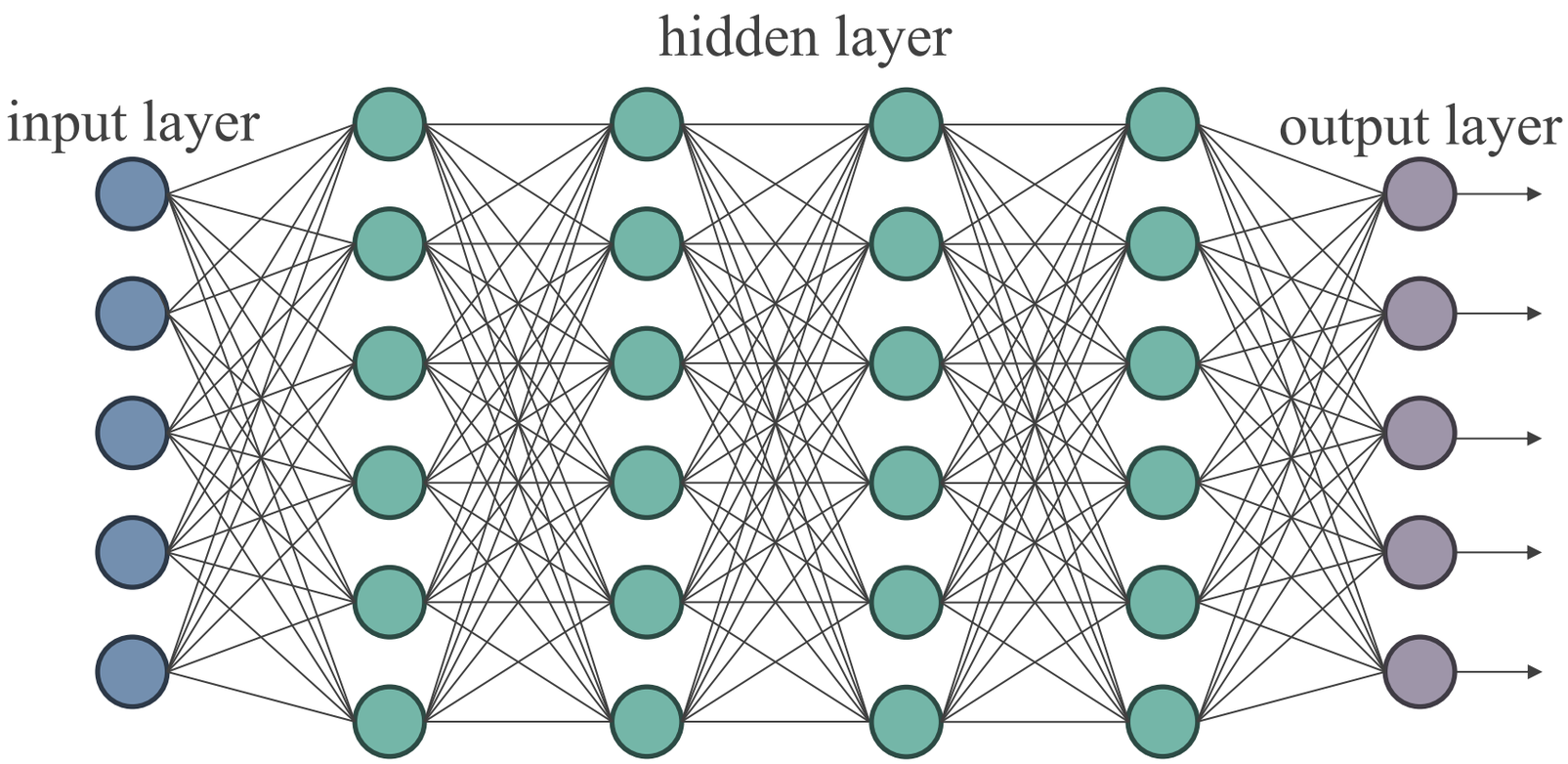
$$\boldsymbol{p} = \boldsymbol{S}(\boldsymbol{U}\,\boldsymbol{G} - \boldsymbol{V}\,\boldsymbol{D})$$

$$\underline{\boldsymbol{G}} \leq \boldsymbol{G} \leq \overline{\boldsymbol{G}}$$

$$\underline{\boldsymbol{p}} \leq \boldsymbol{p} \leq \overline{\boldsymbol{p}}$$



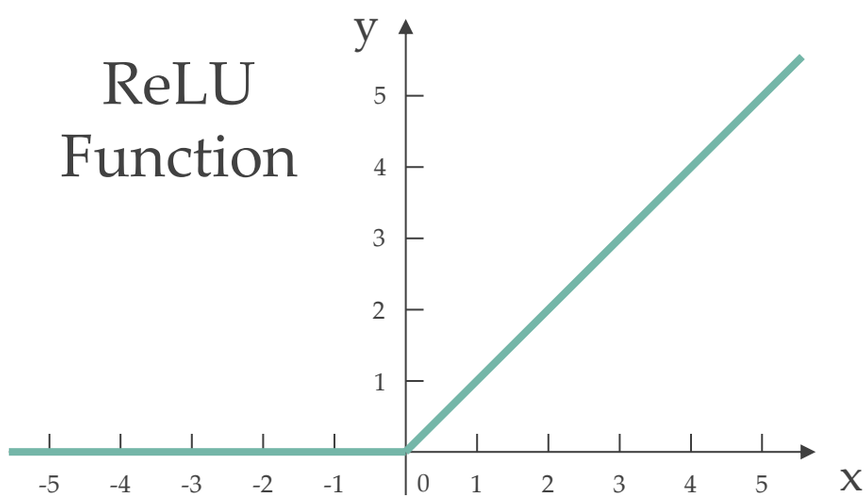
Training Data-based OPF

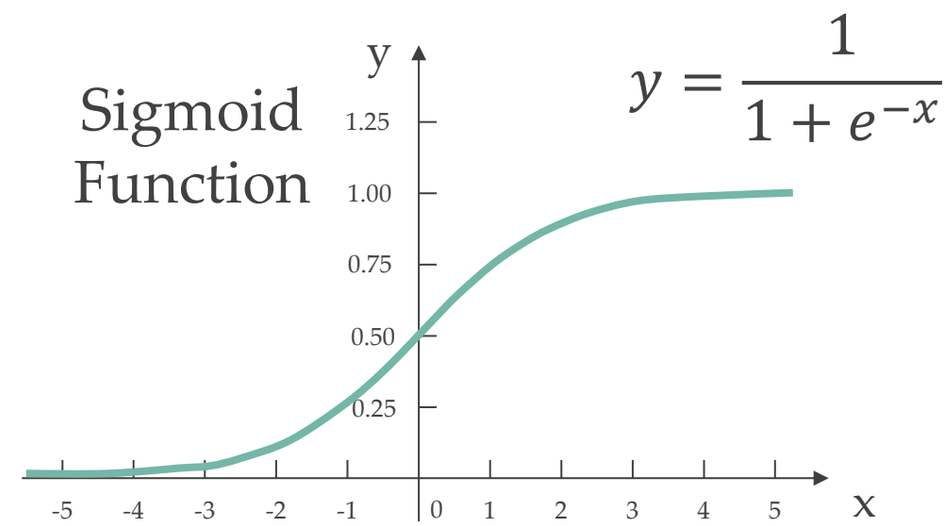


$$z = \sum w_i x_i + b$$

Each neuron is obtained by linear change of all neurons in the previous layer, so neural network can simulate the relationship between inputs and outputs.

Training Data-based OPF





<u>Activation function</u> can add nonlinear parts to the neural network, and with the increase of the depth of the network, the neural network can simulate more complex nonlinear relationships.

Training Data-based OPF

Loss function

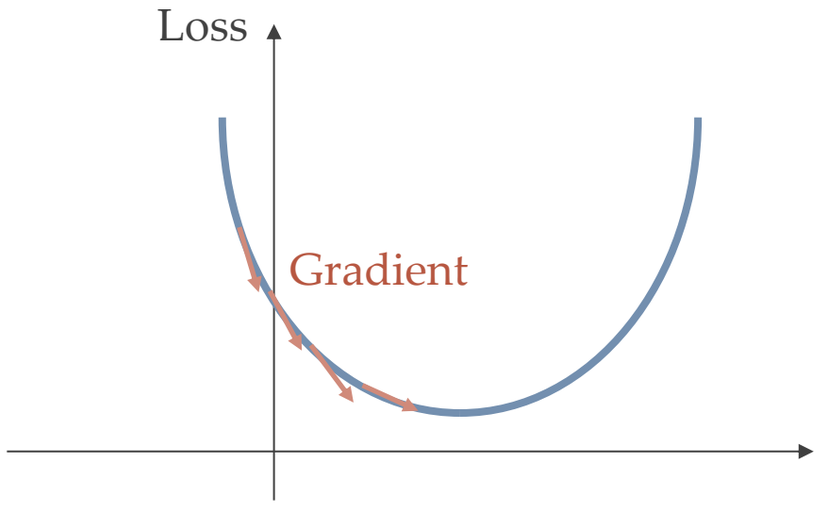
$$L_{loss} = f(y - \hat{y})$$

Loss Function I:

$$L(y, \hat{y}) = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i|$$

Loss Function II:

$$L(y, \hat{y}) = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$$

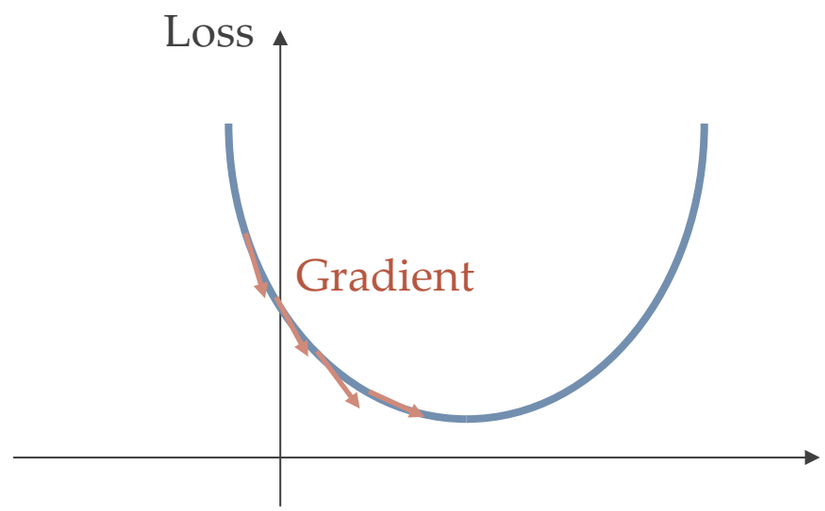


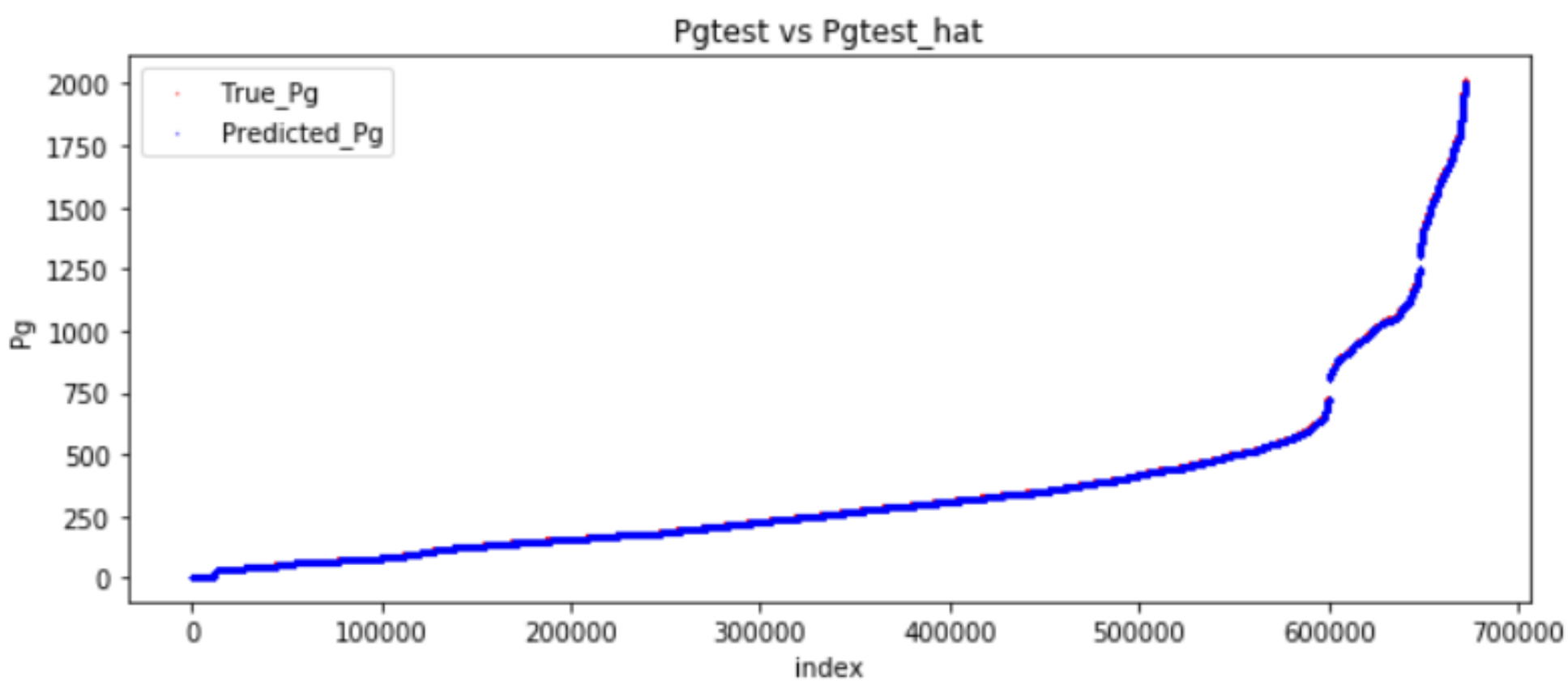
The loss function is used to evaluate the difference between the network output and the target output. By differentiating the loss function, the network parameters w, b can be optimized.

Example 4.5: Training Neural Network for Data-based OPF

Data-/model-based OPF:

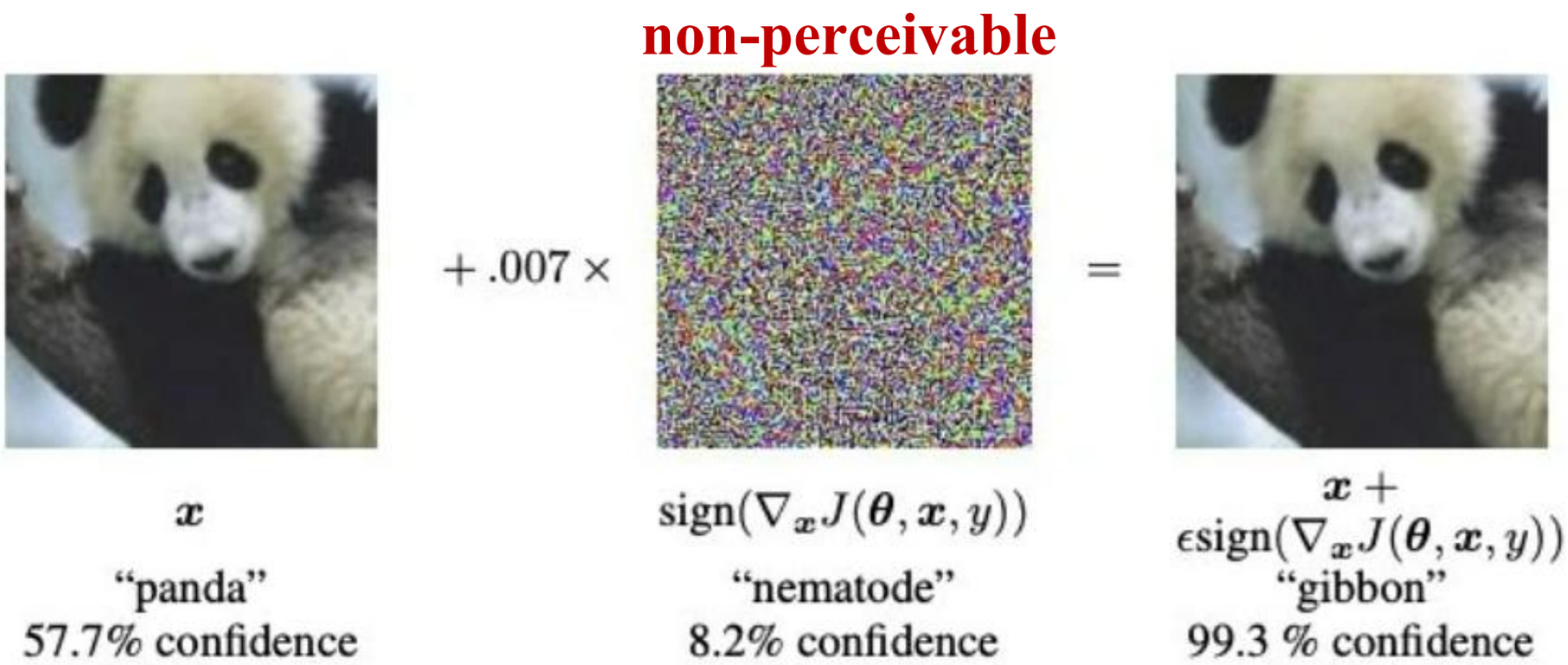
- **train/model**
- **attack**
- **defend**





Attacking Data-based OPF: Fast Gradient Sign Method

- **Adversarial attack** (fast gradient sign method)

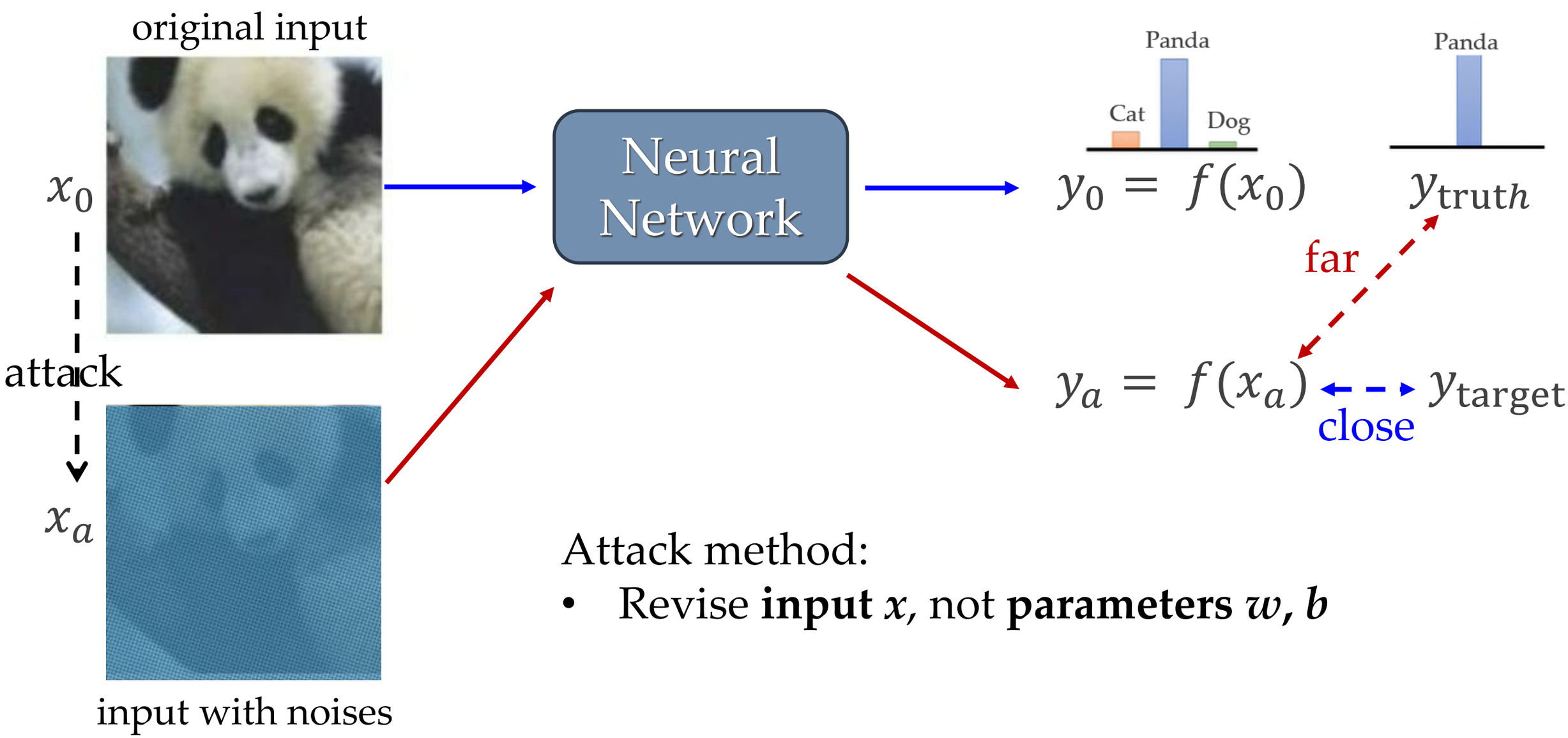


Negligible input Significant output

Goodfellow, Ian J., Jonathon Shlens, and Christian Szegedy. "Explaining and harnessing adversarial examples." *arXiv preprint arXiv:1412.6572* (2014).

Attacking Data-based OPF: Fast Gradient Sign Method

- **Adversarial attack** (fast gradient sign method)



Attack method:

- Revise **input** x, not **parameters** w, b

Goodfellow, Ian J., Jonathon Shlens, and Christian Szegedy. "Explaining and harnessing adversarial examples." *arXiv preprint arXiv:1412.6572* (2014).

Attacking Data-based OPF: Fast Gradient Sign Method

- **Adversarial attack** (fast gradient sign method)

Similar to train neural networks: $x = arg \min_x L_a(x)$

Criticism: Original FGSM attack needs too many costs, and has unclear benefits.
(Rome was not built in a day.)

Attack target/objective:

- Far away from ground truth (FGSM) $L_a(x) = -e(y_a(x) - y_{\text{truth}})$
- Close to some target value (target attack) $L_a(x) = e(y_a(x) - y_{\text{target}})$
- Lower attack cost (negligible attack) $L_a(x) = e(x_a - x_0)$
- Combination of above (max benefit, min cost)

Attacking Data-based OPF: Solution

- **Adversarial attack** (fast gradient sign method) $x = arg\min_{x} L_a(x)$

Gradient descent:

Start from original input x

For t = 1 to T

$$x^t \leftarrow x^{t-1} - \eta \boldsymbol{g}$$

$$\boldsymbol{g} = \begin{bmatrix} \left.\frac{\partial J}{\partial x_1}\right|_{x=x^{t-1}} \\ \left.\frac{\partial J}{\partial x_2}\right|_{x=x^{t-2}} \\ \vdots \end{bmatrix}$$

FGSM:

Start from original input x

For t = 1 ~~to T~~

$$x^t \leftarrow x^{t-1} + \underbrace{\eta}_{\varepsilon} \underbrace{\boldsymbol{g}}_{\begin{bmatrix} +1 \\ -1 \\ +1 \\ \vdots \end{bmatrix}}$$

$$\boldsymbol{g} = \begin{matrix} \pm 1 \\ \pm 1 \\ \\ \end{matrix} \begin{bmatrix} sign(\left.\frac{\partial J}{\partial x_1}\right|_{x=x^{t-1}}) \\ sign(\left.\frac{\partial J}{\partial x_2}\right|_{x=x^{t-2}}) \\ \vdots \end{bmatrix}$$

If $t > 0$, $sign(t) = 1$; otherwise, $sign(t) = -1$

Attacking Data-based OPF: Lower Attack Cost

- **Adversarial attack** (one-pixel attack)

Criticism: Original FGSM attack needs too many costs, and has unclear benefits.
(Rome was not built in a day.)

Su, Jiawei, Danilo Vasconcellos Vargas, and Kouichi Sakurai. "One pixel attack for fooling deep neural networks." *IEEE Transactions on Evolutionary Computation* 23.5 (2019): 828-841.

Attacking Data-based OPF: Lower Attack Cost

- **Adversarial attack** (black/white box attack)

❑ <u>White Box Attack</u>

- Adversaries know the network parameters θ.
- Testing as developer.
- E.g., Fast Gradient Sign Method, Projected Gradient Descent,…

Aleksander Madry, Aleksandar Makelov, Ludwig Schmidt, Dimitris Tsipras, Adrian Vladu. "Towards Deep Learning Models Resistant to Adversarial Attacks." *arXiv preprint arXiv:1706.06083* (2019).

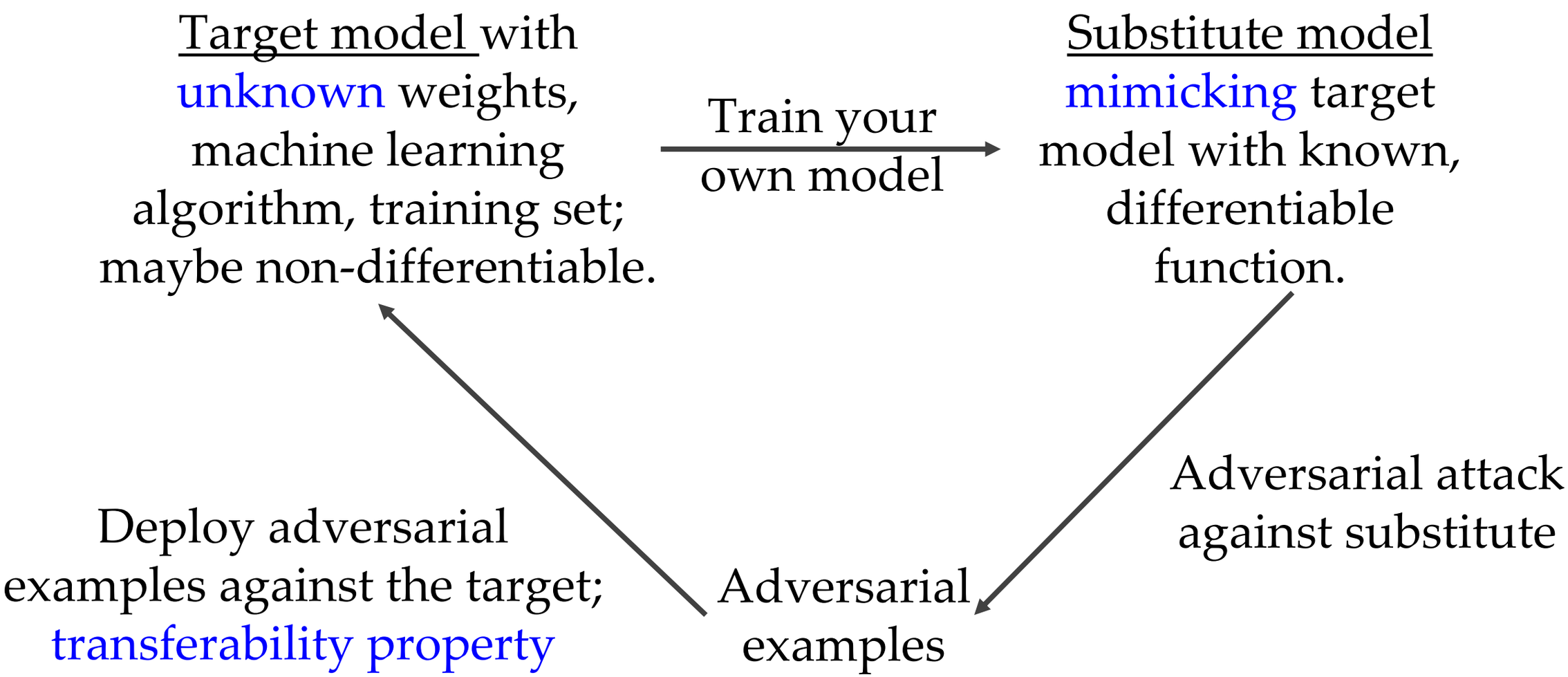
❑ <u>Black Box Attack</u>

- Adversaries do not have access to the model parameters, only model output are observed.
- Testing as user.
- E.g., Transferability Attack, Genetic Algorithm,…

Yanpei Liu, Xinyun Chen, Chang Liu, and Dawn Song. "Delving into transferable adversarial examples and black box attacks. " International Conference on Learning Repre sentations, 2017.

Attacking Data-based OPF: Lower Attack Cost

- **Adversarial attack** (black box attack)



Yanpei Liu, Xinyun Chen, Chang Liu, and Dawn Song. "Delving into transferable adversarial examples and black box attacks. " International Conference on Learning Repre sentations, 2017.

Example 4.6: Attacking Data-based OPF using FGSM

Before FGSM attack:

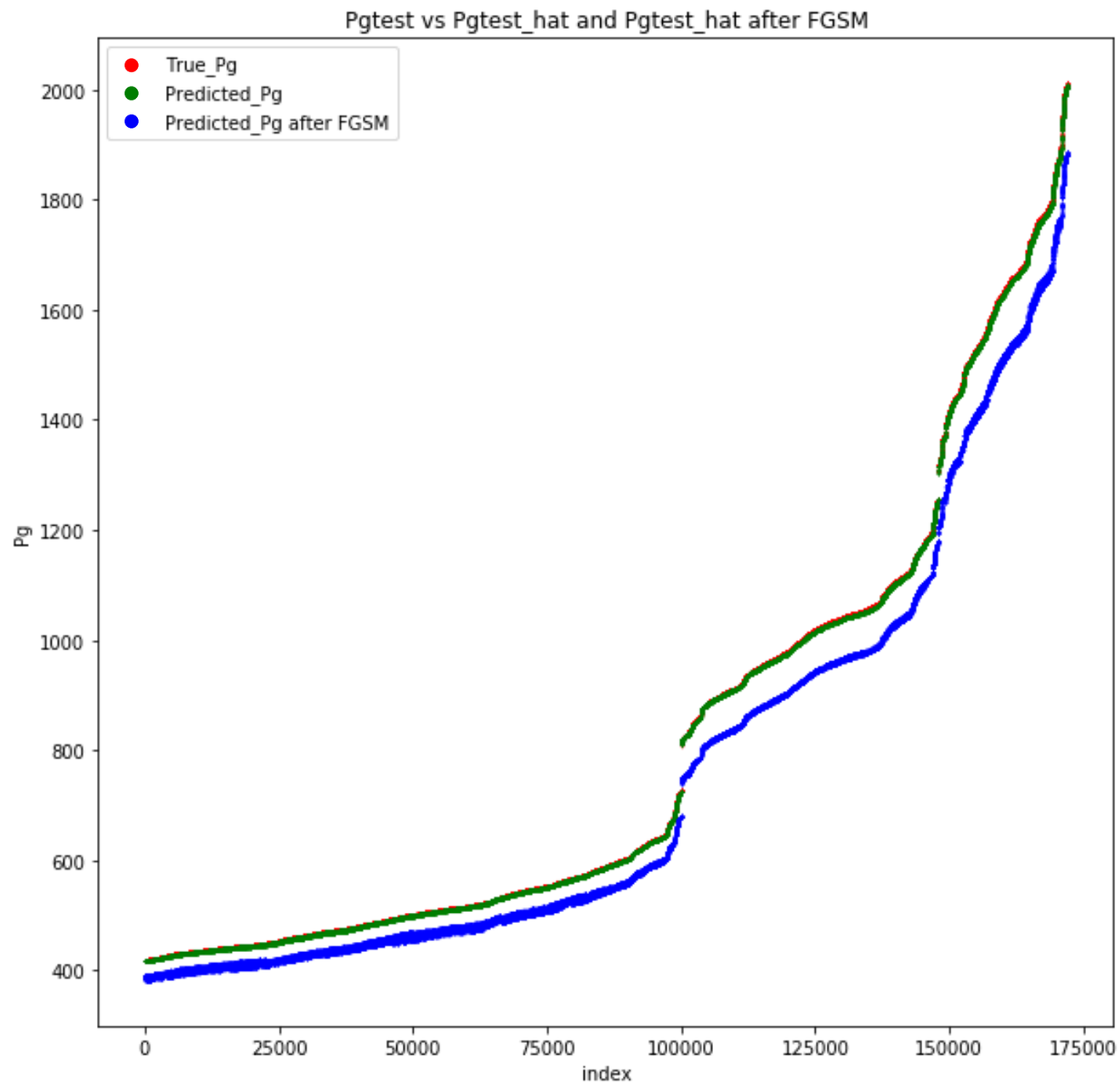
MAE=1.2856
MRE=0.3976

After FGSM attack:

MAE=26.8648
MRE=7.6875

MAE: mean absolute error

MRE: mean relative error

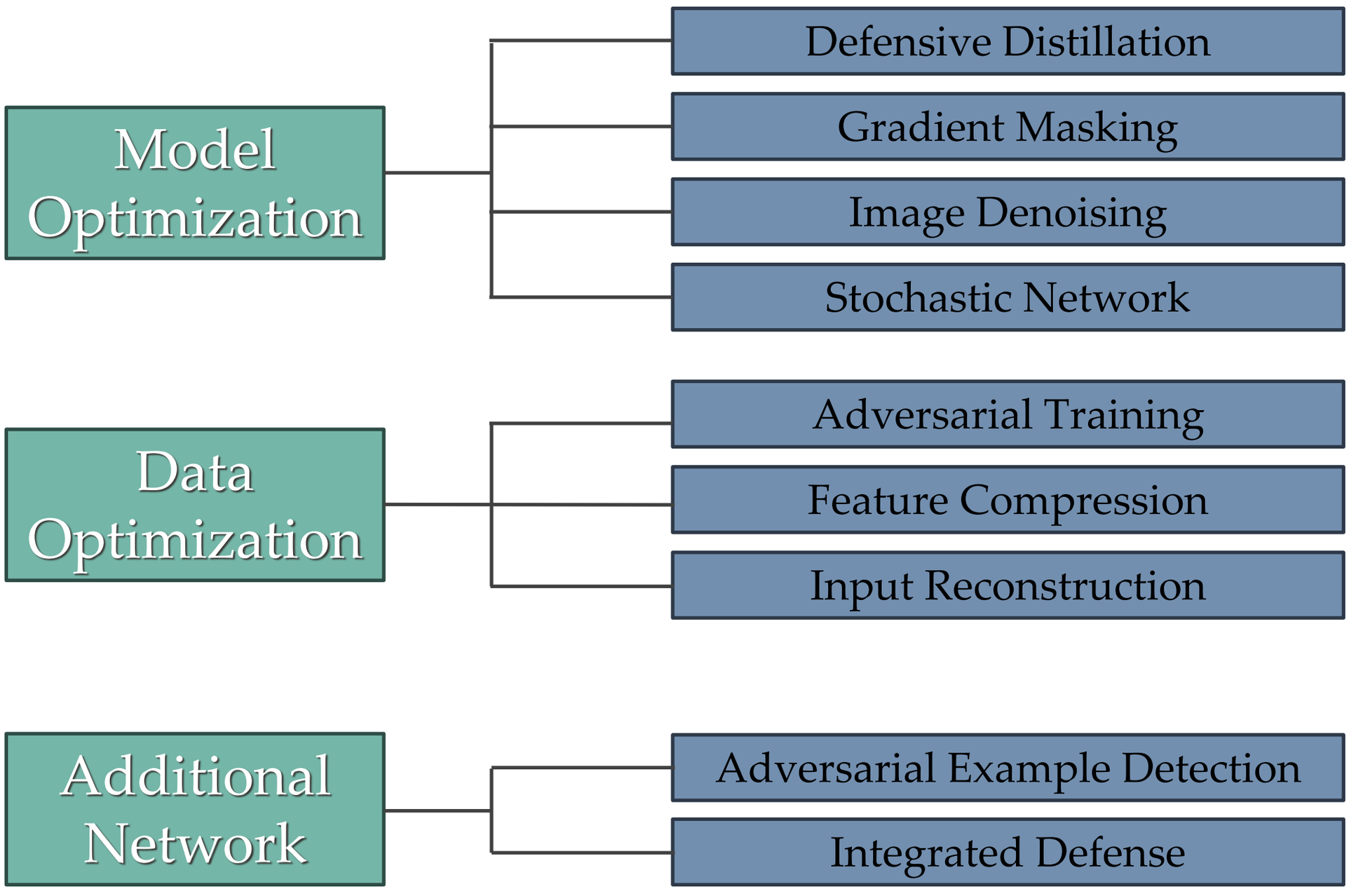


Defending Data-based OPF

Recall cyber-defense methods for model-based OPF:

- spatial- / rule- based strategy
  1. standard largest normalized residual
  2. minimal set of meters $\boldsymbol{z}$
  3. minimal set of state variables $\boldsymbol{x}$ (PMU)
  4. cross validation of extended sensors (based on physical laws)
- temporal- / data- based strategy (rely on training data)
  1. comparison of measurements and predicted values
  2. feature extraction
  3. supervised / unsupervised learning (classification)
  4. hypothesis test (statistical technique)
- active (moving target defense)
- computer science technique (firewall, encryption)
- cyber security technique (industrial security)

Defending Data-based OPF



Defending Data-based OPF: Adversarial Training

Given training set $\mathcal{X} = \{(x^1, y^1), (x^2, y^2), \dots (x^N, y^N)\}$

Using $\mathcal{X}$ to train model

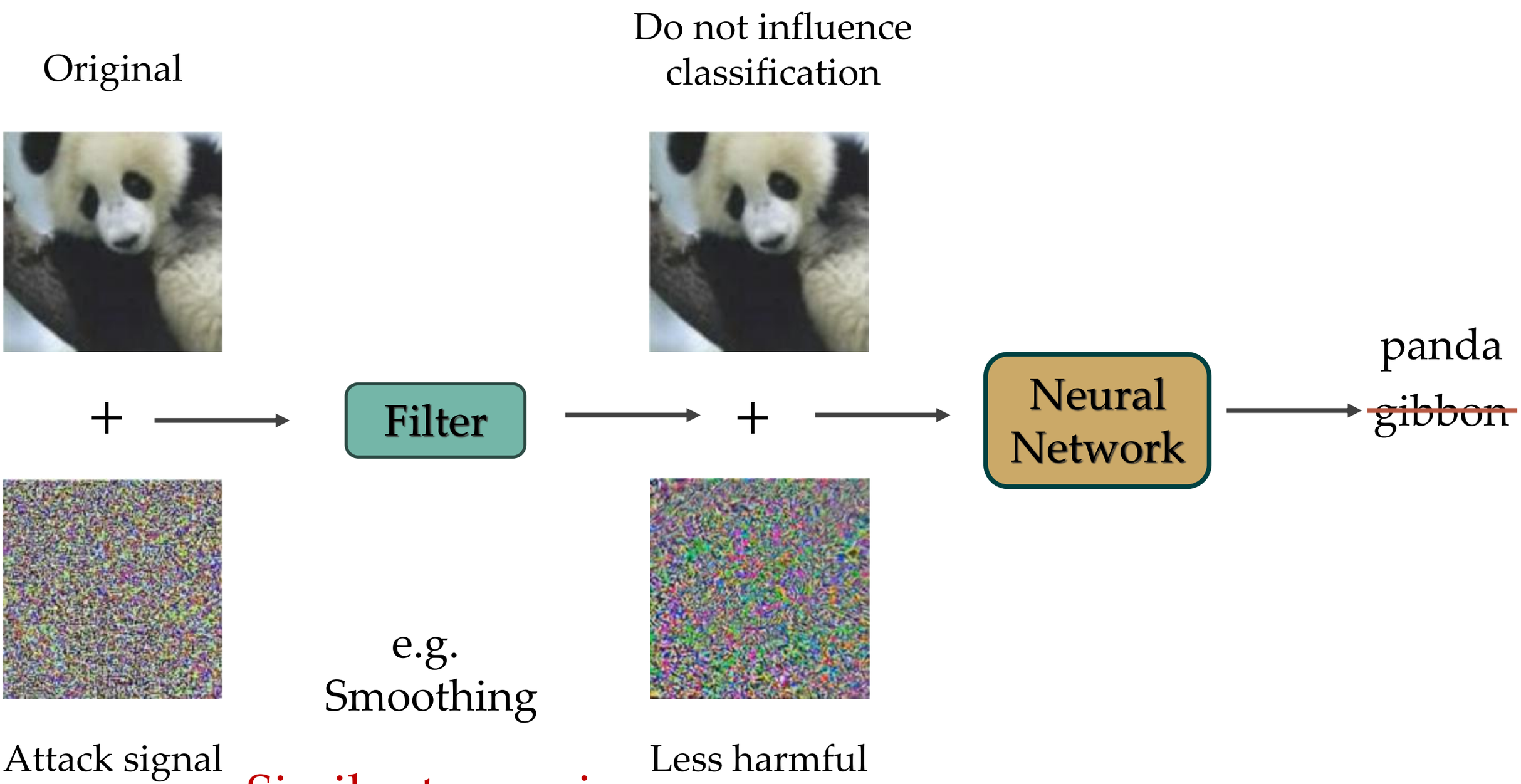
For $n = 1$ to N, Find adversarial input $\bar{x}^n$ given x^n by an attack algorithm

Obtain new training data $\mathcal{X}' = \{(\bar{x}^1, y^1), (\bar{x}^2, y^2), \dots (\bar{x}^N, y^N)\}$

Use both $\mathcal{X}$ and $\mathcal{X}'$ to update the model

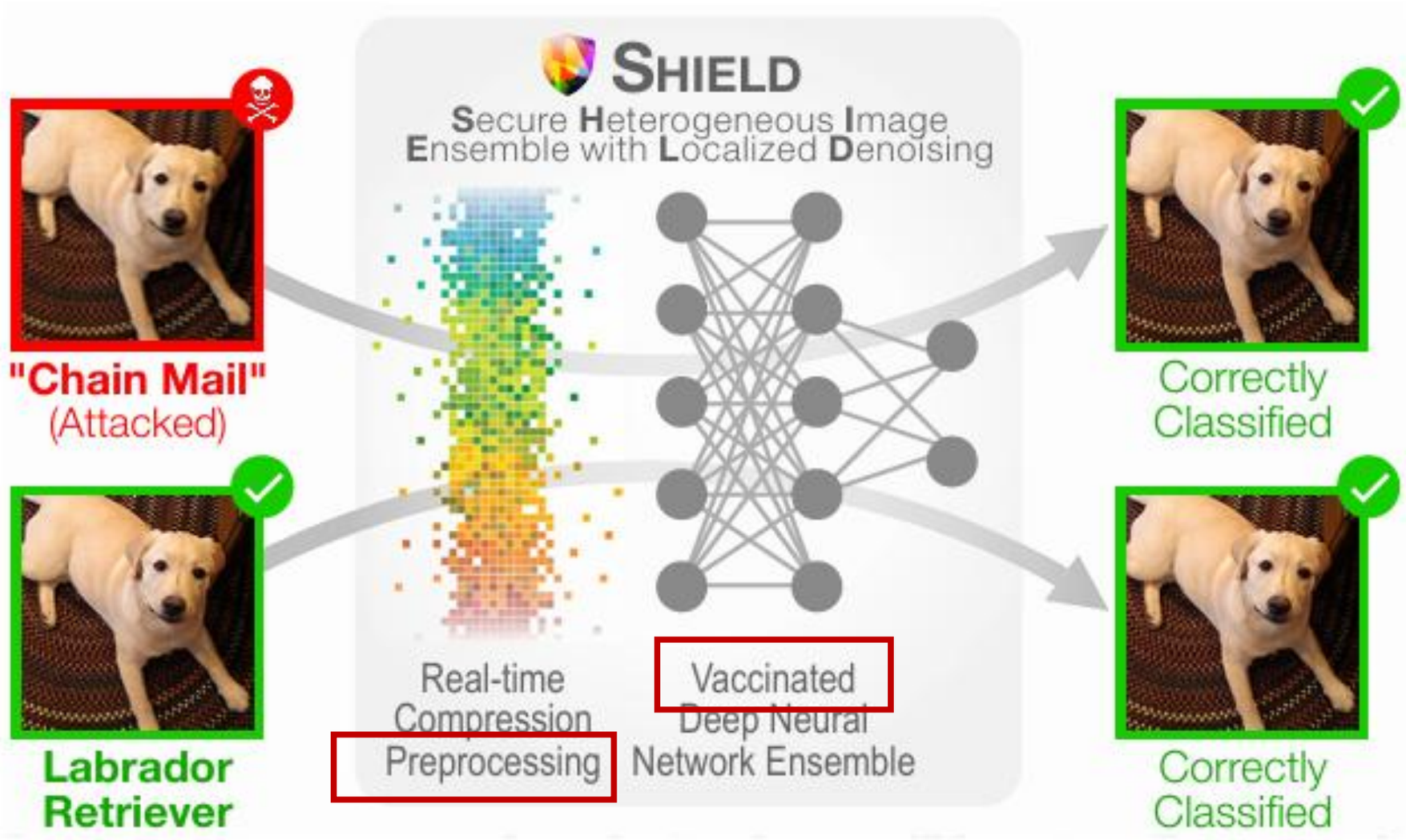
Cyber vaccination!

Defending Data-based OPF: Feature Compression



Similar to moving target defense.

Defending Data-based OPF: Feature Compression



Nilaksh Das, Madhuri Shanbhogue, Shang-Tse Chen, Fred Hohman, Siwei Li, Li Chen, Michael E. Kounavis, Duen Horng Chau. "Shield: Fast, Practical Defense and Vaccination for Deep Learning using JPEG Compression. " arXiv preprint arXiv:1802.06816 (2018).

Data-based OPF: Attack Vs Defense Game

Relying on naive evaluation leads to an arms race that defenders lose

discovery

Szegedy, Christian, Zaremba, Wojciech, Sutskever, Ilya, Bruna, Joan, Erhan, Dumitru, Goodfellow, Ian J., and Fergus, Rob. "Intriguing properties of neural networks. " *arXiv preprint arXiv:1312.6199* (2014).

adversarial training [Goodfellow et al. '15]

Goodfellow, Ian J., Jonathon Shlens, and Christian Szegedy. "Explaining and harnessing adversarial examples." In ICLR, 2015.

transfer attacks

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Outline

- Background
- Attacker Perspective
  (How do cyber-attacks bypass bad data detection?)
- System Perspective
  (What would happen after stealthy cyber-attacks?)
- Defender Perspective
  (How to defend the stealthy cyber-attack?)
- Data-driven Cybersecurity
- **Take-away Message**

Take-away Message: Model-based Attack/Defense

- **Attacker Perspective**
  - Stealthy attack by satisfying power flow equation (Break rules by following rules)
  - Reasonable assumptions of attacking reading/writing capabilities
- **System Perspective**
  - Inappropriate re-dispatch due to OPF that are misled by attacks
  - Manipulation of attack objectives: economy/security/stability/…
- **Defender Perspective**
  - Passive/active cyber-defense strategy
  - Effectiveness and cost-effectiveness

Take-away Message: Data-driven Attack/Defense

- **Training Neural Network**
  - Parallel to mode-based OPF solution, but faster
  - Small change in input, large change in output
- **Adversarial Attack**
  - FGSM as a typical attack method
  - Max attack benefit, min attack cost, trade-off
- **Defense: Adversarial Training**
  - Cyber vaccination and/or preprocessing in advance
  - Effectiveness and cost-effectiveness

Take-away Message: Compare Model-/Data-based Cybersecurity

Model-based OPF:

- **Model** (power flow equation, bad data detection)
- **Attack** (false data injection attack)
- **Defend** (passive/active)

$$\min_{\boldsymbol{G}} \boldsymbol{c}^T \boldsymbol{G}$$

$$\text{s.t.} \quad \mathbf{1}^T \boldsymbol{D} = \mathbf{1}^T \boldsymbol{G}$$

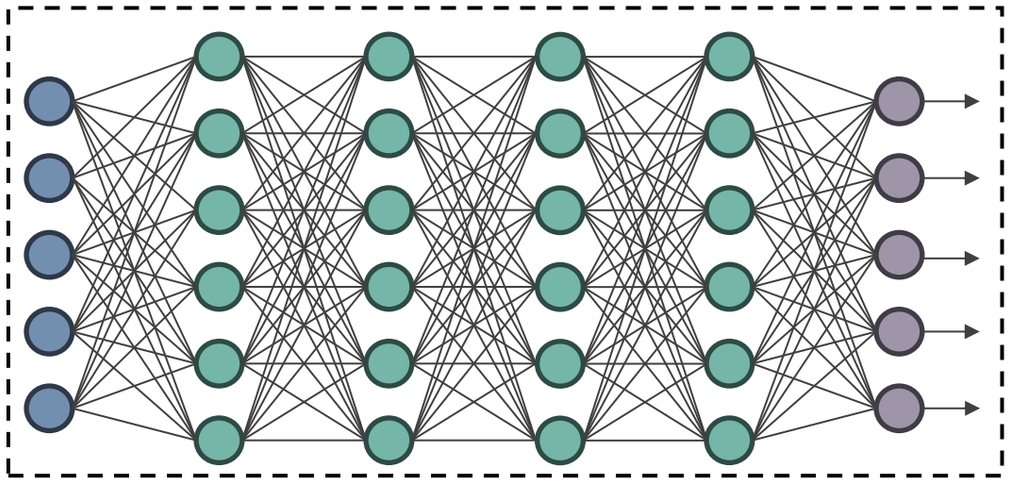
$$\boldsymbol{p} = \boldsymbol{S}(\boldsymbol{U}\,\boldsymbol{G} - \boldsymbol{V}\,\boldsymbol{D})$$

$$\underline{\boldsymbol{G}} \leq \boldsymbol{G} \leq \overline{\boldsymbol{G}}$$

$$\underline{\boldsymbol{p}} \leq \boldsymbol{p} \leq \overline{\boldsymbol{p}}$$

Data-based OPF:

- **Train** (neural network, min loss function)
- **Attack** (adversarial attack)
- **Defend** (passive/active)



Supplementary Reading

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